

```
1 // ===== FILE: index.tsx =====
2 import React from 'react';
3 import ReactDOM from 'react-dom/client';
4 import App from './App';
5 import { ProjectProvider } from './context/ProjectContext';
6
7 const rootElement = document.getElementById('root');
8 if (!rootElement) {
9   throw new Error("Could not find root element to mount to");
10 }
11
12 const root = ReactDOM.createRoot(rootElement);
13 root.render(
14   <React.StrictMode>
15     <ProjectProvider>
16       <App />
17     </ProjectProvider>
18   </React.StrictMode>
19 );
20 // ===== FILE: services/memory.ts =====
21 /**
22  * 记忆管理服务
23  * 管理全局记忆、项目历史、用户偏好等
24  */
25
26 import { openDB, DBSchema, IDBPDatabase } from 'idb';
27 import { ProjectSummary } from './api';
28
29 // ===== IndexedDB Schema =====
30
31 interface ZeroCarbonMemoryDB extends DBSchema {
32   projects: {
33     key: string;
34     value: ProjectSummary;
35     indexes: { 'by-timestamp': string };
36   };
37   preferences: {
38     key: string;
39     value: Record<string, any>;
40   };
41   templates: {
42     key: string;
43     value: Record<string, any>;
44   };
45   learnings: {
46     key: string;
47     value: Record<string, any>;
48   };
49 }
50
```

```
51 // ===== 类型定义 =====
52
53 export interface MemoryData {
54   lastAccessedProject: string | null;
55   recentProjects: string[];
56   preferences: {
57     defaultRegion: string;
58     defaultBuildingType: string;
59     calculationMode: 'simple' | 'advanced';
60     autoSaveInterval: number;
61   };
62   templates: {
63     regionFactors: Record<string, RegionFactor>;
64   };
65   learnings: Record<string, any>;
66 }
67
68 export interface RegionFactor {
69   region: string;
70   climate: {
71     avgTemp: number;
72     tempAmplitude: number;
73     dailyAmplitude: number;
74   };
75   energyPrice: {
76     touTemplate: PriceSegment[];
77     spotPriceTrend: number[];
78   };
79 }
80
81 export interface PriceSegment {
82   start: number;
83   end: number;
84   price: number;
85   type: string;
86 }
87
88 // ===== localStorage 封装 =====
89
90 const STORAGE_PREFIX = 'zce_memory_';
91
92 function get(key: string): string | null {
93   return localStorage.getItem(STORAGE_PREFIX + key);
94 }
95
96 function set(key: string, value: string): void {
97   localStorage.setItem(STORAGE_PREFIX + key, value);
98 }
99
100 function remove(key: string): void {
```

```
101  localStorage.removeItem(STORAGE_PREFIX + key);
102  }
103
104  function parseStored<T>(key: string, fallback: T): T {
105    const raw = get(key);
106    if (!raw) return fallback;
107    try {
108      return JSON.parse(raw) as T;
109    } catch {
110      return fallback;
111    }
112  }
113
114  // ===== 记忆服务类 =====
115
116  class MemoryService {
117    private db: IDBPDatabase<ZeroCarbonMemoryDB> | null = null;
118    private dbName = 'ZeroCarbonMemory';
119    private dbVersion = 1;
120    private isOnline = navigator.onLine;
121
122    constructor() {
123      // 监听网络状态
124      window.addEventListener('online', () => {
125        this.isOnline = true;
126        this.syncPendingData();
127      });
128      window.addEventListener('offline', () => {
129        this.isOnline = false;
130      });
131    }
132
133    // ===== 初始化 =====
134
135    async init(): Promise<void> {
136      if (this.db) return;
137
138      try {
139        this.db = await openDB<ZeroCarbonMemoryDB>({this.dbName, this.dbVersion, {
140          upgrade(db) {
141            // 创建项目存储
142            const projectStore = db.createObjectStore('projects', {
143              keyPath: 'projectId',
144            });
145            projectStore.createIndex('by-timestamp', 'timestamp');
146
147            // 创建偏好设置存储
148            db.createObjectStore('preferences', {
149              keyPath: 'id',
150            });
```

```
151
152     // 创建模板存储
153     db.createObjectStore('templates', {
154         keyPath: 'id',
155     });
156
157     // 创建学习数据存储
158     db.createObjectStore('learnings', {
159         keyPath: 'id',
160     });
161 },
162 });
163 console.log('IndexedDB initialized successfully');
164 } catch (error) {
165     console.error('Failed to initialize IndexedDB:', error);
166     // 降级到localStorage
167 }
168 }
169
170 // ===== 快速记忆 (localStorage) =====
171
172 getQuickMemory(): MemoryData {
173     const stored = parseStored<Partial<MemoryData>>>('memory', {});
174     const storedPreferences: Partial<MemoryData['preferences']> = stored.preferences || {};
175     return {
176         lastAccessedProject: get('lastAccessedProject') || stored.lastAccessedProject || null,
177         recentProjects: parseStored<string[]>('recentProjects', stored.recentProjects || []),
178         preferences: {
179             defaultRegion: parseStored<string>('pref_defaultRegion', storedPreferences.defaultR...
180             defaultBuildingType: parseStored<string>('pref_defaultBuildingType', storedPreferen...
181             calculationMode: parseStored<'simple' | 'advanced'>('pref_calculationMode', storedP...
182             autoSaveInterval: parseStored<number>('pref_autoSaveInterval', storedPreferences.au...
183         },
184         templates: {
185             regionFactors: parseStored<Record<string, RegionFactor>>('regionFactors', stored.te...
186         },
187         learnings: parseStored<Record<string, any>>('learnings', stored.learnings || {}),
188     };
189 }
190
191 setQuickMemory(key: keyof MemoryData, value: any): void {
192     const memory = this.getQuickMemory();
193     memory[key] = value;
194     set('memory', JSON.stringify(memory));
195     if (key === 'lastAccessedProject') {
196         set('lastAccessedProject', value || '');
197     } else if (key === 'recentProjects' || key === 'learnings') {
198         set(key, JSON.stringify(value));
199     } else if (key === 'templates') {
200         set('regionFactors', JSON.stringify(value?.regionFactors || {}));
```

```
201  }
202  }
203
204  // ===== 项目管理 =====
205
206  async saveProject(project: ProjectSummary): Promise<void> {
207    // 保存到IndexedDB
208    if (this.db) {
209      const tx = this.db.transaction('projects', 'readwrite');
210      await tx.store.put({
211        ...project,
212        savedAt: new Date().toISOString(),
213      });
214    }
215
216    // 更新最近项目列表
217    await this.addRecentProject(project.projectId, project.projectName);
218  }
219
220  async getProject(projectId: string): Promise<ProjectSummary | null> {
221    if (!this.db) return null;
222
223    const tx = this.db.transaction('projects', 'readonly');
224    return await tx.store.get(projectId);
225  }
226
227  async getAllProjects(): Promise<ProjectSummary[]> {
228    if (!this.db) return [];
229
230    const tx = this.db.transaction('projects', 'readonly');
231    return await tx.store.getAll();
232  }
233
234  async deleteProject(projectId: string): Promise<void> {
235    // 从IndexedDB删除
236    if (this.db) {
237      const tx = this.db.transaction('projects', 'readwrite');
238      await tx.store.delete(projectId);
239    }
240
241    // 从最近列表中移除
242    const memory = this.getQuickMemory();
243    memory.recentProjects = memory.recentProjects.filter(id => id !== projectId);
244    this.setQuickMemory('recentProjects', memory.recentProjects);
245
246    // 如果是当前项目，清空
247    if (memory.lastAccessedProject === projectId) {
248      this.setQuickMemory('lastAccessedProject', null);
249    }
250  }
```

```
251
252 // ===== 最近项目 =====
253
254 async addRecentProject(projectId: string, projectName: string): Promise<void> {
255   const memory = this.getQuickMemory();
256   const recent = memory.recentProjects.filter(id => id !== projectId);
257   recent.unshift(projectId);
258   memory.recentProjects = recent.slice(0, 10); // 最多10个
259   memory.lastAccessedProject = projectId;
260
261   set('recentProjects', JSON.stringify(memory.recentProjects));
262   set('lastAccessedProject', projectId);
263   set('memory', JSON.stringify(memory));
264 }
265
266 getRecentProjects(): string[] {
267   return this.getQuickMemory().recentProjects;
268 }
269
270 getLastAccessedProject(): string | null {
271   return get('lastAccessedProject') || null;
272 }
273
274 // ===== 用户偏好 =====
275
276 setPreference(key: string, value: any): void {
277   const memory = this.getQuickMemory();
278   memory.preferences[key] = value;
279   set('memory', JSON.stringify(memory));
280
281   // 单独存储以便快速访问
282   set(`pref_${key}`, JSON.stringify(value));
283 }
284
285 getPreference<T>(key: string): T | null {
286   const memory = this.getQuickMemory();
287   const value = memory.preferences[key];
288   if (value !== undefined) return value as T;
289
290   // 从单独存储读取
291   const stored = get(`pref_${key}`);
292   return stored ? JSON.parse(stored) : null;
293 }
294
295 // ===== 区域知识库 =====
296
297 getRegionFactors(): Record<string, RegionFactor> {
298   return this.getQuickMemory().templates.regionFactors;
299 }
300
```

```
301 getRegionFactor(region: string): RegionFactor | null {
302   const factors = this.getRegionFactors();
303   return factors[region] || null;
304 }
305
306 async saveRegionFactor(region: string, factor: RegionFactor): Promise<void> {
307   const memory = this.getQuickMemory();
308   memory.templates.regionFactors[region] = factor;
309   set('regionFactors', JSON.stringify(memory.templates.regionFactors));
310   set('memory', JSON.stringify(memory));
311
312   // 同时保存到IndexedDB
313   if (this.db) {
314     const tx = this.db.transaction('templates', 'readwrite');
315     await tx.store.put({ id: `region_${region}`, data: factor });
316   }
317 }
318
319 // ===== 学习数据 =====
320
321 async updateLearnings(data: Record<string, any>): Promise<void> {
322   const memory = this.getQuickMemory();
323   memory.learnings = { ...memory.learnings, ...data };
324   set('learnings', JSON.stringify(memory.learnings));
325   set('memory', JSON.stringify(memory));
326
327   // 同时保存到IndexedDB
328   if (this.db) {
329     const tx = this.db.transaction('learnings', 'readwrite');
330     for (const [key, value] of Object.entries(data)) {
331       await tx.store.put({ id: key, data: value });
332     }
333   }
334 }
335
336 getLearnings(): Record<string, any> {
337   return this.getQuickMemory().learnings;
338 }
339
340 // ===== 智能建议 =====
341
342 getSmartSuggestions(currentProject: any): Array<{
343   type: string;
344   module: string;
345   message: string;
346 }> {
347   const suggestions: Array<{ type: string; module: string; message: string }> = [];
348   const learnings = this.getLearnings();
349   const memory = this.getQuickMemory();
350
```

```
351 // 基于区域特征的建议
352 const region = currentProject?.projectBaseInfo?.province;
353 if (region) {
354   const regionFactor = this.getRegionFactor(region);
355   if (regionFactor) {
356     // 检查是否需要应用电价模板
357     if (!memory.preferences.defaultRegion || memory.preferences.defaultRegion !== regio...
358       suggestions.push({
359         type: 'configuration',
360         module: 'price',
361         message: `检测到${region}地区，已应用对应的电价模板`,
362       });
363     }
364   }
365 }
366
367 // 基于历史项目的建议
368 const recent = memory.recentProjects.slice(0, 5);
369 if (recent.length > 0) {
370   // 检查储能配置
371   const currentStorageRatio =
372     currentProject?.modules?.['retrofit-storage']?.investment /
373     (currentProject?.totalInvestment || 1);
374   if (currentStorageRatio < 0.15) {
375     suggestions.push({
376       type: 'investment',
377       module: 'storage',
378       message: '储能配置偏低，建议配置15%以上容量以提高峰谷套利收益',
379     });
380   }
381 }
382
383 return suggestions;
384 }
385
386 // ===== 清理 =====
387
388 async clearOldProjects(daysOld: number = 90): Promise<void> {
389   if (!this.db) return;
390
391   const cutoffDate = new Date();
392   cutoffDate.setDate(cutoffDate.getDate() - daysOld);
393
394   const tx = this.db.transaction(['projects', 'readwrite']);
395   const projects = await tx.store.getAll();
396   const idsToDelete = projects
397     .filter(p => new Date(p.timestamp) < cutoffDate)
398     .map(p => p.projectId);
399
400   for (const id of idsToDelete) {
```



```
401   await tx.store.delete(id);
402 }
403
404 console.log(`Cleared ${idsToDelete.length} old projects`);
405 }
406
407 async clearAllData(): Promise<void> {
408   // 清空localStorage
409   Object.keys(localStorage)
410     .filter(key => key.startsWith(STORAGE_PREFIX))
411     .forEach(key => localStorage.removeItem(key));
412
413   // 清空IndexedDB
414   if (this.db) {
415     const stores = ['projects', 'preferences', 'templates', 'learnings'];
416     for (const storeName of stores) {
417       const tx = this.db.transaction(storeName, 'readwrite');
418       await tx.store.clear();
419     }
420   }
421 }
422
423 // ===== 同步 =====
424
425 private async syncPendingData(): Promise<void> {
426   // 重新上线时同步数据
427   console.log('Syncing pending data...');
428   // 这里可以实现与后端的数据同步逻辑
429 }
430
431 // ===== 工具方法 =====
432
433 getStorageUsage(): { used: number; total: number } {
434   // 计算localStorage使用量
435   let used = 0;
436   Object.keys(localStorage)
437     .filter(key => key.startsWith(STORAGE_PREFIX))
438     .forEach(key => {
439       used += localStorage.getItem(key)?.length || 0;
440     });
441
442   return {
443     used,
444     total: 5 * 1024 * 1024, // 约5MB
445   };
446 }
447 }
448
449 // 导出单例
450 export const memoryService = new MemoryService();
```

```
451 // ===== FILE: utils/financial.ts =====
452 /**
453  * 零碳项目统一财务工具
454  * 提供标准 IRR (内部收益率) 计算及现金流推演逻辑
455  */
456
457 /**
458  * 标准 IRR 计算 (牛顿迭代法)
459  * @param cashFlows 现金流数组 [第0年投资(负数), 第1年收益, ..., 第N年收益]
460  * @param guess 初始推测值 (默认 0.1)
461  * @returns 内部收益率 (百分比)
462  */
463 export function calculateIRR(cashFlows: number[], guess: number = 0.1): number {
464   if (cashFlows.length < 2) return 0;
465
466   // 检查是否有投资(负数)和收益(正数)
467   const hasNegative = cashFlows.some(v => v < 0);
468   const hasPositive = cashFlows.some(v => v > 0);
469   if (!hasNegative || !hasPositive) return 0;
470
471   let period = cashFlows.length;
472   let rate = guess;
473
474   for (let i = 0; i < 50; i++) {
475     let npv = 0;
476     let dnpv = 0;
477
478     for (let t = 0; t < period; t++) {
479       const factor = Math.pow(1 + rate, t);
480       npv += cashFlows[t] / factor;
481       if (t > 0) {
482         dnpv -= t * cashFlows[t] / (factor * (1 + rate));
483       }
484     }
485
486     if (Math.abs(npv) < 1e-4) return rate * 100;
487     if (Math.abs(dnpv) < 1e-10) break; // 防止除以 0
488
489     const newRate = rate - npv / dnpv;
490     if (Math.abs(newRate - rate) < 1e-6) return newRate * 100;
491
492     rate = newRate;
493
494     // 边界检查: 防止 IRR 跑到非理性区间
495     if (rate > 5 || rate < -0.99) break;
496   }
497
498   // 如果牛顿法失败, 尝试二分法兜底
499   return calculateIRRBisection(cashFlows);
500 }
```

```
501
502 function calculateIRRBisection(cashFlows: number[]): number {
503     let low = -0.99;
504     let high = 5.0;
505
506     for (let i = 0; i < 40; i++) {
507         let mid = (low + high) / 2;
508         let npv = 0;
509         for (let t = 0; t < cashFlows.length; t++) {
510             npv += cashFlows[t] / Math.pow(1 + mid, t);
511         }
512
513         if (Math.abs(npv) < 1e-4) return mid * 100;
514         if (npv > 0) low = mid;
515         else high = mid;
516     }
517     return low * 100;
518 }
519
520 /**
521  * 生成标准 25 年现金流
522  * 完全贴合“增值税留抵结转”及“附加税减半”小微企业模型
523  */
524 export function generateStandardCashFlows(params: {
525     totalInvestment: number,
526     totalGrossSaving: number, // 每年毛收益 (含税)
527     omRate: number, // 年度常规运维费率 (%)
528     insuranceRate?: number, // 保险费率 (%)
529     taxRate: number, // 企业所得税率 (%) (小微默认5%)
530     vatRate?: number, // 增值税率 (%) (默认13%)
531     vatExtraRate?: number, // 附加税费率 (%) (城建税7%+教育附加3%+地方教育附加2% = 12%, 小微减半后为6%)
532     degradationLinear?: number, // 发电量每年线性衰减率 (%)
533     degradationFirstYear?: number, // 首年衰减 (%)
534     period?: number // 计算年限, 默认 25
535 }): number[]) {
536     const {
537         totalInvestment,
538         totalGrossSaving,
539         omRate,
540         insuranceRate = 0,
541         taxRate,
542         vatRate = 13.0,
543         vatExtraRate = 6.0, // 贴合小微企业“六税两费”减半政策, 默认为6%
544         degradationLinear = 0.45,
545         degradationFirstYear = 1.0,
546         period = 25
547     } = params;
548
549     const cashFlows: number[] = [-totalInvestment];
550
```

```
551 // 运维、保险总费用(含税)
552 const totalOpexRate = omRate + insuranceRate;
553 const annualOpexGross = totalInvestment * (totalOpexRate / 100);
554 // 运维费进项税抵扣(由于电站运维常含人工及其他服务,假设综合抵扣率为5%并倒算,或者直接假设运维包含对应进项)
555 // 根据提供的表格,“运维、保险等进项税”约占固定值。此处为保证逻辑完备,假设运维费用含6%进项税。
556 const opexVatRate = 0.05; // 参考图片的进项率
557 const opexInputVat = annualOpexGross / (1 + opexVatRate) * opexVatRate;
558 const annualOpexNet = annualOpexGross - opexInputVat;
559
560 // 初始投资进项税一次性留抵(固定资产抵扣)
561 const investmentNet = totalInvestment / (1 + vatRate / 100);
562 let vatCarryForward = totalInvestment - investmentNet; // 第一期建站产生的进项税总额
563
564 let accumulatedDepreciation = 0;
565
566 for (let year = 1; year <= period; year++) {
567     // 计算产量衰减
568     let degradationFactor = 1.0;
569     if (year === 1) {
570         degradationFactor = 1 - (degradationFirstYear / 100);
571     } else {
572         degradationFactor = (1 - (degradationFirstYear / 100)) * Math.pow(1 - (degradat...
573     }
574
575     // 年度发电含税收入
576     const currentYearSavingGross = totalGrossSaving * degradationFactor;
577
578     // 1. 增值税计算(应交增值税 = 销项税 - 进项税 - 上期留抵)
579     const revenueNet = currentYearSavingGross / (1 + vatRate / 100);
580     const outputVat = currentYearSavingGross - revenueNet;
581
582     // 增值税额推演(允许负数留抵)
583     let vatPayable = outputVat - opexInputVat - vatCarryForward;
584
585     if (vatPayable < 0) {
586         vatCarryForward = -vatPayable; // 转入下期留抵
587         vatPayable = 0; // 本期无需缴纳增值税
588     } else {
589         vatCarryForward = 0; // 留抵扣完
590     }
591
592     // 2. 税金及附加(基于本期实际缴纳的增值税额计算,享有减半优惠则为6%)
593     const vatSurcharge = vatPayable * (vatExtraRate / 100);
594
595     // 3. 企业所得税(利润额 = 不含税收入 - 不含税运维 - 折旧 - 附加税费)
596     // 10年双倍余额递减法折旧(Double Declining Balance)
597     let currentYearDepreciation = 0;
598     if (year <= 8) {
599         currentYearDepreciation = (investmentNet - accumulatedDepreciation) * (2 / 10);
600         accumulatedDepreciation += currentYearDepreciation;
```

```
601     } else if (year === 9 || year === 10) {
602         const remainingForYear8 = investmentNet - accumulatedDepreciation;
603         currentYearDepreciation = remainingForYear8 / 2; // 平摊剩余净值
604         // 不再累加，避免第10年结束后不等于总额（实则第10年提完已是全额）
605     }
606
607     const taxableIncome = revenueNet - annualOpexNet - currentYearDepreciation - vatSur...
608
609     // 所得税抵扣
610     const incomeTax = taxableIncome > 0 ? taxableIncome * (taxRate / 100) : 0;
611
612     // 4. 当年净现金流（含税收入 - 含税支出 - 交纳增值税 - 附加税 - 所得税）
613     const netCashFlow = currentYearSavingGross - annualOpexGross - vatPayable - vatSurc...
614
615     cashFlows.push(parseFloat(netCashFlow.toFixed(4)));
616 }
617
618 return cashFlows;
619 }
620
621 /**
622  * 计算静态/动态回本周期
623  * @param cashFlows 现金流数组 [第0年投资(负数), 第1年收益, ..., 第N年收益]
624  * @returns 回本年数(带小数)
625  */
626 export function calculatePaybackPeriod(cashFlows: number[]): number {
627     if (cashFlows.length === 0 || cashFlows[0] >= 0) return 0;
628
629     let cumulative = cashFlows[0];
630     for (let i = 1; i < cashFlows.length; i++) {
631         const prevCumulative = cumulative;
632         cumulative += cashFlows[i];
633
634         if (cumulative >= 0) {
635             // 线性插值计算具体月份
636             return (i - 1) + (Math.abs(prevCumulative) / cashFlows[i]);
637         }
638     }
639
640     return cashFlows.length - 1; // 未回本返回总年限
641 }
642 // ===== FILE: utils/reportExport.ts =====
643 /**
644  * PDF和Word报告导出工具函数
645  * 用于生成完整的汇报报告
646  */
647
648 import { PRODUCT_IDENTITY } from '../shared/config/productIdentity';
649
650 const REPORT_GENERATOR = `${PRODUCT_IDENTITY.fullName}${PRODUCT_IDENTITY.version}`;
```

```
651
652 function formatLocalDateCompact(date = new Date()): string {
653   return [date.getFullYear(), String(date.getMonth() + 1).padStart(2, '0'), String(date.get...
654 }
655
656 // ===== PDF导出功能 =====
657
658 /**
659  * 打印/导出为PDF
660  * 使用浏览器原生打印功能，用户可以选择"保存为PDF"
661  */
662 export function exportToPDF(options?: {
663   title?: string;
664   filename?: string;
665 }) {
666   // 保存原始标题
667   const originalTitle = document.title;
668
669   // 设置打印标题
670   if (options?.title) {
671     document.title = options.title;
672   }
673
674   // 调用打印对话框
675   window.print();
676
677   // 恢复原始标题
678   setTimeout(() => {
679     document.title = originalTitle;
680   }, 100);
681 }
682
683 /**
684  * 生成打印样式的HTML内容
685  */
686 export function generatePrintableHTML(content: {
687   projectInfo: any;
688   modules: any[];
689   financial: any;
690   charts?: any[];
691 }): string {
692   const { projectInfo, modules, financial, charts } = content;
693   const activeInvestment = modules.filter(m => m.isActive).reduce((sum, module) => sum + mo...
694   const activeAnnualBenefit = modules.filter(m => m.isActive).reduce((sum, module) => sum +...
695   const horizonYears = financial.period ?? 25;
696
697   return `
698 <!DOCTYPE html>
699 <html>
700 <head>
```

```
701 <meta charset="utf-8">
702 <title>${projectInfo.name} - 项目估值报告</title>
703 <style>
704 @page {
705     size: A4 portrait;
706     margin: 1.5cm;
707 }
708
709 @media print {
710     body {
711         -webkit-print-color-adjust: exact;
712         print-color-adjust: exact;
713     }
714     @page {
715         margin: 1.5cm;
716         size: A4 portrait;
717     }
718     /* 允许内容在需要时自动分页 */
719     * {
720         page-break-inside: auto !important;
721         page-break-after: auto !important;
722         page-break-before: auto !important;
723     }
724     /* 表格行在需要时可以跨页 */
725     tr, td, th {
726         page-break-inside: auto !important;
727     }
728 }
729
730 body {
731     font-family: 'Noto Sans SC', 'Microsoft YaHei', sans-serif;
732     line-height: 1.4;
733     color: #333;
734     max-width: 210mm;
735     margin: 0 auto;
736     padding: 0;
737 }
738
739 .header {
740     text-align: center;
741     margin-bottom: 20px;
742     border-bottom: 2px solid #4f46e5;
743     padding-bottom: 10px;
744 }
745
746 .header h1 {
747     font-size: 22px;
748     margin: 0 0 6px 0;
749     color: #4f46e5;
750 }
```

```
751
752 .header p {
753     margin: 3px 0;
754     color: #666;
755     font-size: 12px;
756 }
757
758 .section {
759     margin-top: 30px;
760     margin-bottom: 20px;
761     padding-top: 15px;
762 }
763
764 .section h2 {
765     font-size: 16px;
766     color: #1e293b;
767     border-left: 3px solid #4f46e5;
768     padding-left: 10px;
769     margin-bottom: 10px;
770     margin-top: 0;
771 }
772
773 .info-table {
774     width: 100%;
775     border-collapse: collapse;
776     margin-bottom: 12px;
777 }
778
779 .info-table th,
780 .info-table td {
781     border: 1px solid #e2e8f0;
782     padding: 6px 8px;
783     text-align: left;
784     font-size: 12px;
785 }
786
787 .info-table th {
788     background-color: #f8fafc;
789     font-weight: 600;
790     width: 30%;
791 }
792
793 .data-table {
794     width: 100%;
795     border-collapse: collapse;
796     margin-bottom: 12px;
797 }
798
799 .data-table th,
800 .data-table td {
```



```
801     border: 1px solid #e2e8f0;
802     padding: 5px 6px;
803     text-align: center;
804     font-size: 11px;
805 }
806
807 .data-table th {
808     background-color: #4f46e5;
809     color: white;
810     font-weight: 600;
811     font-size: 11px;
812 }
813
814 .data-table tr:nth-child(even) {
815     background-color: #f8f9fc;
816 }
817
818 .data-table .active {
819     background-color: #ecfdf5 !important;
820 }
821
822 .kpi-grid {
823     display: grid;
824     grid-template-columns: repeat(4, 1fr);
825     gap: 8px;
826     margin-bottom: 12px;
827 }
828
829 .kpi-card {
830     background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
831     color: white;
832     padding: 12px;
833     border-radius: 6px;
834     text-align: center;
835     min-height: 80px;
836 }
837
838 .kpi-card .label {
839     font-size: 10px;
840     opacity: 0.9;
841     margin-bottom: 5px;
842 }
843
844 .kpi-card .value {
845     font-size: 18px;
846     font-weight: bold;
847 }
848
849 .chart-placeholder {
850     background: #f8f9fc;
```

```
851     border: 2px dashed #cbd5e1;
852     border-radius: 8px;
853     padding: 30px;
854     text-align: center;
855     color: #64748b;
856 }
857
858 .footer {
859     text-align: center;
860     margin-top: 25px;
861     padding-top: 12px;
862     border-top: 1px solid #e2e8f0;
863     color: #64748b;
864     font-size: 10px;
865 }
866 </style>
867 </head>
868 <body>
869 <!-- 封面 -->
870 <div class="header">
871     <h1>零碳项目收益估值报告</h1>
872     <p><strong>项目名称 : </strong>${projectInfo.name}</p>
873     <p><strong>项目类型 : </strong>${projectInfo.type}</p>
874     <p><strong>生成时间 : </strong>${new Date().toLocaleString('zh-CN')}</p>
875 </div>
876
877 <!-- 项目基本信息 -->
878 <div class="section">
879     <h2>一、项目基本信息</h2>
880     <table class="info-table">
881         <tr><th>项目名称</th><td>${projectInfo.name}</td></tr>
882         <tr><th>项目类型</th><td>${projectInfo.type}</td></tr>
883         <tr><th>所在地区</th><td>${projectInfo.province} ${projectInfo.city}</td></tr>
884         <tr><th>建筑数量</th><td>${projectInfo.buildings?.length || 0} 栋</td></tr>
885     </table>
886 </div>
887
888 <!-- 改造方案概览 -->
889 <div class="section">
890     <h2>二、改造方案概览</h2>
891     <table class="data-table">
892         <thead>
893             <tr>
894                 <th>序号</th>
895                 <th>模块名称</th>
896                 <th>状态</th>
897                 <th>策略</th>
898                 <th>投资额(万元)</th>
899                 <th>年收益(万元)</th>
900                 <th>ROI(%)</th>
```

```
901     </tr>
902   </thead>
903   <tbody>
904     ${modules.map((m, i) => `
905       <tr class="${m.isActive ? 'active' : ''}">
906         <td>${i + 1}</td>
907         <td>${m.name}</td>
908         <td>${m.isActive ? '启用' : '禁用'}</td>
909         <td>${m.strategy}</td>
910         <td>${m.investment.toFixed(2)}</td>
911         <td>${m.yearlySaving.toFixed(2)}</td>
912         <td>${m.investment > 0 ? `((${m.yearlySaving / m.investment} * 100).toFixed(1))...
913       </tr>
914     `).join("")}
915     <tr style="font-weight: bold; background-color: #e0e7ff;">
916       <td colspan="4">合计</td>
917       <td>${activeInvestment.toFixed(2)}</td>
918       <td>${activeAnnualBenefit.toFixed(2)}</td>
919       <td>${activeInvestment > 0 ? `((${activeAnnualBenefit / activeInvestment} * 100)....
920     </tr>
921   </tbody>
922 </table>
923 </div>
924
925 <!-- 财务分析 -->
926 <div class="section">
927   <h2>三、财务综合分析</h2>
928   <div class="kpi-grid">
929     <div class="kpi-card">
930       <div class="label">初始总投资</div>
931       <div class="value">¥${financial.totalInvestment.toFixed(1)}万</div>
932     </div>
933     <div class="kpi-card">
934       <div class="label">净现值 (NPV)</div>
935       <div class="value">¥${financial.npv.toFixed(1)}万</div>
936     </div>
937     <div class="kpi-card">
938       <div class="label">内部收益率 (IRR)</div>
939       <div class="value">${financial.irr.toFixed(2)}%</div>
940     </div>
941     <div class="kpi-card">
942       <div class="label">回本周期</div>
943       <div class="value">${financial.payback > horizonYears ? `>${horizonYears}` : financ...
944     </div>
945   </div>
946 </div>
947
948 ${financial.participants?.length ? `
949 <div class="section">
950   <h2>四、参与方财务账本</h2>
```

```
951 <table class="data-table">
952   <thead><tr><th>参与方</th><th>承担投资(万元)</th><th>首年净收益(万元)</th><th>NPV(万元)</th><th>IRR</th>...
953   <tbody>${financial.participants.map((p: any) => `<tr><td>${p.name}</td><td>${p.invest...
954 </table>
955 </div>`;"}
956
957 <!-- 页脚 -->
958 <div class="footer">
959   <p>本报告由${REPORT_GENERATOR}自动生成</p>
960   <p>生成时间: ${new Date().toLocaleString('zh-CN')}</p>
961 </div>
962 </body>
963 </html>
964 `;
965 }
966
967 /**
968  * 生成可打印的HTML文档并导出
969  */
970 export function generateAndPrintReport(content: {
971   projectInfo: any;
972   modules: any[];
973   financial: any;
974 }): void {
975   const html = generatePrintableHTML(content);
976
977   // 创建新窗口
978   const printWindow = window.open("", '_blank');
979   if (!printWindow) {
980     alert('请允许弹窗以打印报告');
981     return;
982   }
983
984   // 写入HTML内容
985   printWindow.document.write(html);
986   printWindow.document.close();
987
988   // 等待内容加载后打印
989   setTimeout(() => {
990     printWindow.focus();
991     printWindow.print();
992   }, 500);
993 }
994
995 // ===== Word导出功能 =====
996
997 /**
998  * 生成Word文档并下载
999  * 使用HTML格式，Word可以直接打开
1000  */
```

```
1001 export function exportToWord(content: {
1002   projectInfo: any;
1003   modules: any[];
1004   financial: any;
1005   options?: {
1006     includeCharts?: boolean;
1007   };
1008 }): void {
1009   const { projectInfo, modules, financial, options = {} } = content;
1010   const horizonYears = financial.period ?? 25;
1011   const moduleChapterHtml = (financial.moduleDetails || []).filter((item: any) => ['retrofi...
1012   let cumulative = 0;
1013   return `
1014     <div style="page-break-before: always;"></div>
1015     <h2>${item.name}完整方案</h2>
1016     <div class="kpi-box"><div class="kpi-value">¥${item.investment.toFixed(2)}万</div><div...
1017     <div class="kpi-box"><div class="kpi-value">¥${item.firstYearNetBenefit.toFixed(2)}万<...
1018     <div class="kpi-box"><div class="kpi-value">${item.irr.toFixed(2)}%</div><div class="...
1019     <div class="kpi-box"><div class="kpi-value">${item.payback.toFixed(2)}年</div><div cla...
1020     <h3>方案与技术参数</h3>
1021     <table><tr><th>参数</th><th>数值</th></tr>${item.technicalMetrics.map((metric: any) => `<...
1022     <h3>25年独立现金流</h3>
1023     <table><tr><th>年份</th><th>年度净现金流(万元)</th><th>累计现金流(万元)</th></tr>${item.cashFlows.map(...
1024     <h3>假设与风险</h3>
1025     <ul>${[...item.assumptions, ...item.warnings].map((text: string) => `<li>${text}</li>...
1026   `;
1027   }).join("");
1028
1029   const wordContent = `
1030   <html xmlns:o="urn:schemas-microsoft-com:office:office" xmlns:w="urn:schemas-microsoft-com:...
1031   <head>
1032     <meta charset="utf-8">
1033     <title>${projectInfo.name} - 项目估值报告</title>
1034     <style>
1035       body { font-family: "Microsoft YaHei", sans-serif; line-height: 1.6; margin: 40px; }
1036       h1 { color: #4f46e5; border-bottom: 3px solid #4f46e5; padding-bottom: 10px; }
1037       h2 { color: #1e293b; border-left: 4px solid #4f46e5; padding-left: 12px; margin-top: 30...
1038       h3 { color: #334155; margin-top: 22px; }
1039       table { border-collapse: collapse; width: 100%; margin: 20px 0; }
1040       th, td { border: 1px solid #e2e8f0; padding: 10px; text-align: center; }
1041       th { background-color: #4f46e5; color: white; }
1042       tr:nth-child(even) { background-color: #f8fafc; }
1043       .info-row { margin: 10px 0; }
1044       .info-label { font-weight: bold; display: inline-block; width: 120px; }
1045       .kpi-box { background: #f0f9ff; padding: 15px; margin: 10px; display: inline-block; wid...
1046       .kpi-value { font-size: 24px; font-weight: bold; color: #4f46e5; }
1047       .kpi-label { font-size: 12px; color: #64748b; }
1048     </style>
1049   </head>
1050   <body>
```

```
1051 <h1>零碳项目收益估值报告</h1>
1052
1053 <h2>一、项目基本信息</h2>
1054 <div class="info-row"><span class="info-label">项目名称 : </span>${projectInfo.name}</div>
1055 <div class="info-row"><span class="info-label">项目类型 : </span>${projectInfo.type}</div>
1056 <div class="info-row"><span class="info-label">所在地区 : </span>${projectInfo.province} ${proj...
1057 <div class="info-row"><span class="info-label">生成时间 : </span>${new Date().toLocaleString('z...
1058
1059 <h2>二、改造方案概览</h2>
1060 <table>
1061   <tr><th>序号</th><th>模块名称</th><th>状态</th><th>投资额(万元)</th><th>年收益(万元)</th><th>ROI(%)</th><th><...
1062   ${modules.map((m, i) => `
1063     <tr style="${m.isActive ? 'background-color: #ecfdf5'; ': '}">
1064       <td>${i + 1}</td>
1065       <td>${m.name}</td>
1066       <td>${m.isActive ? '启用' : '禁用'}</td>
1067       <td>${m.investment.toFixed(2)}</td>
1068       <td>${m.yearlySaving.toFixed(2)}</td>
1069       <td>${m.investment > 0 ? ` ${(m.yearlySaving / m.investment) * 100}.toFixed(1)}%` : ...
1070     </tr>
1071   `).join("")}
1072 </table>
1073
1074 ${moduleChapterHtml}
1075
1076 <div style="page-break-before: always;"></div>
1077 <h2>光储联合与系统财务汇总</h2>
1078 <div class="kpi-box"><div class="kpi-value">¥${financial.totalInvestment.toFixed(1)}万</di...
1079 <div class="kpi-box"><div class="kpi-value">¥${financial.npv.toFixed(1)}万</div><div class...
1080 <div class="kpi-box"><div class="kpi-value">${financial.irr.toFixed(2)}%</div><div class=...
1081 <div class="kpi-box"><div class="kpi-value">${financial.payback > horizonYears ? ` ${hori...
1082
1083 ${financial.interactions?.length ? `<h3>板块协同收益</h3><table><tr><th>协同项目</th><th>年增量价值(万元)<...
1084
1085 ${financial.annualData?.length ? `<h3>系统25年现金流</h3><table><tr><th>年份</th><th>年度净现金流(万元)</...
1086
1087 ${financial.participants?.length ? `
1088 <h2>四、参与方财务账本</h2>
1089 <table>
1090   <tr><th>参与方</th><th>承担投资(万元)</th><th>首年净收益(万元)</th><th>NPV(万元)</th><th>IRR</th><th>回收期<...
1091   ${financial.participants.map((p: any) => `<tr><td>${p.name}</td><td>${p.investment.toFi...
1092 </table>` : ''}
1093
1094 ${financial.assumptions?.length ? `<h3>统一测算依据与风险提示</h3><ul>${financial.assumptions.map((i...
1095
1096 <p style="margin-top: 50px; color: #64748b; font-size: 12px; text-align: center;">
1097   本报告由${REPORT_GENERATOR}自动生成<br/>
1098   生成时间 : ${new Date().toLocaleString('zh-CN')}
1099 </p>
1100 </body>
```

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```
1151 | ${i + 1} | ${m.name} | ${m.isActive ? "": ""} | ${m.investment.toFixed(2)} | ${m.yearl...
1152 | | **合计** | | ${activeInvestment.toFixed(2)} | ${activeAnnualBenefit.toFixed(2)} | ${activ...
1153
1154 ## 财务综合分析
1155
1156 ### 核心指标
1157
1158 - **初始总投资**: ${financial.totalInvestment.toFixed(1)}万
1159 - **净现值 (NPV)**: ${financial.npv.toFixed(1)}万
1160 - **内部收益率 (IRR)**: ${financial.irr.toFixed(2)}%
1161 - **动态回本周期**: ${financial.payback > horizonYears ? `>${horizonYears}` : financial.payback.t...
1162
1163 ---
1164
1165 *本报告由${REPORT_GENERATOR}自动生成*
1166 `;
1167 }
1168
1169 /**
1170  * 导出Markdown报告
1171  */
1172 export function exportMarkdownReport(content: {
1173   projectInfo: any;
1174   modules: any[];
1175   financial: any;
1176 }): void {
1177   const markdown = exportToMarkdown(content);
1178
1179   const blob = new Blob([markdown], {
1180     type: 'text/markdown;charset=utf-8'
1181   });
1182
1183   const link = document.createElement('a');
1184   const url = URL.createObjectURL(blob);
1185   const dateStr = formatLocalDateCompact();
1186   const safeName = content.projectInfo.name.replace(/[\\\/:*?"<>|]/g, '_');
1187
1188   link.href = url;
1189   link.download = `${safeName}_估值报告_${dateStr}.md`;
1190   document.body.appendChild(link);
1191   link.click();
1192   document.body.removeChild(link);
1193   URL.revokeObjectURL(url);
1194 }
1195 // ===== FILE: components/ProjectEntry.tsx =====
1196 import React, { useState, useEffect, useMemo, useCallback, useRef } from 'react';
1197 import { useProject, Transformer } from '../context/ProjectContext';
1198 import { Building, EquipmentItem } from './project-entry/types';
1199 import { BuildingList } from './project-entry/BuildingList';
1200 import { SystemConfigTable } from './project-entry/SystemConfigTable';
```



```
1201 import { BillImport } from './project-entry/BillImport';
1202 import { RestaurantLoadProfilePreview } from './project-entry/RestaurantLoadProfilePreview';
1203 import { VillaLoadProfilePreview } from './project-entry/VillaLoadProfilePreview';
1204 import { LocationPicker } from './LocationPicker';
1205 // -- 省份城市数据 --
1206 const PROVINCE_CITIES: Record<string, string[]> = {
1207     // 直辖市
1208     'Beijing': ['东城区', '西城区', '朝阳区', '海淀区', '丰台区', '石景山区', '通州区', '顺义区', '昌平区', '大兴区'],
1209     'Shanghai': ['浦东新区', '闵行区', '徐汇区', '长宁区', '静安区', '普陀区', '虹口区', '杨浦区', '黄浦区', '宝山区', '嘉定...
1210     'Tianjin': ['和平区', '河东区', '河西区', '南开区', '河北区', '红桥区', '东丽区', '西青区', '津南区', '北辰区', '武清区'...
1211     'Chongqing': ['渝中区', '江北区', '南岸区', '九龙坡区', '沙坪坝区', '大渡口区', '渝北区', '巴南区', '北碚区', '万州区', '...
1212
1213     // 华东地区
1214     'Jiangsu': ['南京市', '苏州市', '无锡市', '常州市', '南通市', '扬州市', '徐州市', '盐城市', '镇江市', '泰州市', '淮安市'...
1215     'Zhejiang': ['杭州市', '宁波市', '温州市', '嘉兴市', '湖州市', '绍兴市', '金华市', '衢州市', '舟山市', '台州市', '丽水市'],
1216     'Shandong': ['济南市', '青岛市', '烟台市', '潍坊市', '临沂市', '淄博市', '济宁市', '泰安市', '威海市', '德州市', '东营市'...
1217     'Anhui': ['合肥市', '芜湖市', '蚌埠市', '淮南市', '马鞍山市', '淮北市', '铜陵市', '安庆市', '黄山市', '滁州市', '阜阳市'....
1218     'Fujian': ['福州市', '厦门市', '泉州市', '漳州市', '莆田市', '三明市', '南平市', '龙岩市', '宁德市'],
1219     'Jiangxi': ['南昌市', '赣州市', '宜春市', '吉安市', '上饶市', '抚州市', '九江市', '景德镇市', '萍乡市', '新余市', '鹰潭市'],
1220
1221     // 华南地区
1222     'Guangdong': ['广州市', '深圳市', '佛山市', '东莞市', '珠海市', '中山市', '惠州市', '江门市', '汕头市', '湛江市', '茂名...
1223     'Guangxi': ['南宁市', '柳州市', '桂林市', '梧州市', '北海市', '防城港市', '钦州市', '贵港市', '玉林市', '百色市', '贺州市'...
1224     'Hainan': ['海口市', '三亚市', '三沙市', '儋州市', '五指山市', '琼海市', '文昌市', '万宁市', '东方市'],
1225
1226     // 华中地区
1227     'Henan': ['郑州市', '洛阳市', '开封市', '南阳市', '许昌市', '平顶山市', '安阳市', '鹤壁市', '新乡市', '焦作市', '濮阳市'....
1228     'Hubei': ['武汉市', '宜昌市', '襄阳市', '荆州市', '十堰市', '黄石市', '孝感市', '荆门市', '鄂州市', '黄冈市', '咸宁市', '...
1229     'Hunan': ['长沙市', '株洲市', '湘潭市', '衡阳市', '邵阳市', '岳阳市', '常德市', '张家界市', '益阳市', '郴州市', '永州市'...
1230
1231     // 华北地区
1232     'Hebei': ['石家庄市', '唐山市', '秦皇岛市', '邯郸市', '邢台市', '保定市', '张家口市', '承德市', '沧州市', '廊坊市', '衡水市'],
1233     'Shanxi': ['太原市', '大同市', '阳泉市', '长治市', '晋城市', '朔州市', '晋中市', '运城市', '忻州市', '临汾市', '吕梁市'],
1234     'Neimenggu': ['呼和浩特市', '包头市', '乌海市', '赤峰市', '通辽市', '鄂尔多斯市', '呼伦贝尔市', '巴彦淖尔市', '乌兰察布市', '...
1235
1236     // 东北地区
1237     'Liaoning': ['沈阳市', '大连市', '鞍山市', '抚顺市', '本溪市', '丹东市', '锦州市', '营口市', '阜新市', '辽阳市', '盘锦市'...
1238     'Jilin': ['长春市', '吉林市', '四平市', '辽源市', '通化市', '白山市', '松原市', '白城市', '延边州'],
1239     'Heilongjiang': ['哈尔滨市', '齐齐哈尔市', '鸡西市', '鹤岗市', '双鸭山市', '大庆市', '伊春市', '佳木斯市', '七台河市', '...
1240
1241     // 西北地区
1242     'Shaanxi': ['西安市', '铜川市', '宝鸡市', '咸阳市', '渭南市', '延安市', '汉中市', '榆林市', '安康市', '商洛市'],
1243     'Gansu': ['兰州市', '嘉峪关市', '金昌市', '白银市', '天水市', '武威市', '张掖市', '平凉市', '酒泉市', '庆阳市', '定西市'....
1244     'Qinghai': ['西宁市', '海东市', '海北州', '黄南州', '海南州', '果洛州', '玉树州', '海西州'],
1245     'Ningxia': ['银川市', '石嘴山市', '吴忠市', '固原市', '中卫市'],
1246     'Xinjiang': ['乌鲁木齐市', '克拉玛依市', '吐鲁番市', '哈密市', '昌吉州', '博尔塔拉州', '巴音郭楞州', '阿克苏地区', '克孜勒苏州'...
```

```
1251   'Yunnan': ['昆明市', '曲靖市', '玉溪市', '保山市', '昭通市', '丽江市', '普洱市', '临沧市', '楚雄州', '红河州', '文山州'],...
1252   'Xizang': ['拉萨市', '日喀则市', '昌都市', '林芝市', '山南市', '那曲市', '阿里地区']
1253   };
1254
1255   const PROVINCE_NAMES: Record<string, string> = {
1256     // 直辖市
1257     'Beijing': '北京市',
1258     'Shanghai': '上海市',
1259     'Tianjin': '天津市',
1260     'Chongqing': '重庆市',
1261     // 华东地区
1262     'Jiangsu': '江苏省',
1263     'Zhejiang': '浙江省',
1264     'Shandong': '山东省',
1265     'Anhui': '安徽省',
1266     'Fujian': '福建省',
1267     'Jiangxi': '江西省',
1268     // 华南地区
1269     'Guangdong': '广东省',
1270     'Guangxi': '广西壮族自治区',
1271     'Hainan': '海南省',
1272     // 华中地区
1273     'Henan': '河南省',
1274     'Hubei': '湖北省',
1275     'Hunan': '湖南省',
1276     // 华北地区
1277     'Hebei': '河北省',
1278     'Shanxi': '山西省',
1279     'Neimenggu': '内蒙古自治区',
1280     // 东北地区
1281     'Liaoning': '辽宁省',
1282     'Jilin': '吉林省',
1283     'Heilongjiang': '黑龙江省',
1284     // 西北地区
1285     'Shaanxi': '陕西省',
1286     'Gansu': '甘肃省',
1287     'Qinghai': '青海省',
1288     'Ningxia': '宁夏回族自治区',
1289     'Xinjiang': '新疆维吾尔自治区',
1290     // 西南地区
1291     'Sichuan': '四川省',
1292     'Guizhou': '贵州省',
1293     'Yunnan': '云南省',
1294     'Xizang': '西藏自治区'
1295   };
1296
1297   // 省份到地区的映射（用于学校类型判断寒暑假天数）
1298   const PROVINCE_REGION: Record<string, 'south' | 'central' | 'north'> = {
1299     // 南方地区（寒暑假较短）
1300     'Shanghai': 'south',
```

```
1301  'Zhejiang': 'south',
1302  'Fujian': 'south',
1303  'Jiangxi': 'south',
1304  'Guangdong': 'south',
1305  'Guangxi': 'south',
1306  'Hainan': 'south',
1307  'Chongqing': 'south',
1308  'Sichuan': 'south',
1309  'Guizhou': 'south',
1310  'Yunnan': 'south',
1311  'Hunan': 'south',
1312  // 中部地区
1313  'Jiangsu': 'central',
1314  'Anhui': 'central',
1315  'Hubei': 'central',
1316  'Henan': 'central',
1317  'Shaanxi': 'central',
1318  // 北方地区（寒假较长）
1319  'Beijing': 'north',
1320  'Tianjin': 'north',
1321  'Hebei': 'north',
1322  'Shanxi': 'north',
1323  'Neimenggu': 'north',
1324  'Liaoning': 'north',
1325  'Jilin': 'north',
1326  'Heilongjiang': 'north',
1327  'Shandong': 'north',
1328  'Gansu': 'north',
1329  'Qinghai': 'north',
1330  'Ningxia': 'north',
1331  'Xinjiang': 'north',
1332  'Xizang': 'north'
1333  };
1334
1335  const defaultBuildings: Building[] = [
1336  {
1337      id: 1,
1338      name: '1号生产车间',
1339      type: '工厂 (Factory)',
1340      area: 12500,
1341      systems: {
1342          lighting: {
1343              enabled: true,
1344              mode: 'density',
1345              density: 12,
1346              items: [
1347                  { id: 1, name: '金卤灯 (High Bay)', power: 250, count: 200 },
1348                  { id: 2, name: 'T8 荧光灯管', power: 36, count: 500 }
1349              ]
1350          },
```

```
1351     hvac: {
1352         enabled: true,
1353         items: [
1354             { id: 1, name: '1# 离心冷水机组', type: '水冷冷水机组 (Chiller)', health: '良 (5-10年...',
1355             { id: 2, name: '组合式空调箱 (AHU)', type: '风冷模块机组', health: '优 (1-5年)', powe...
1356         ]
1357     },
1358     water: {
1359         enabled: false,
1360         items: [
1361             { id: 1, name: '燃气热水锅炉', type: 'Boiler', power: 200, count: 1 }
1362         ]
1363     },
1364     production: {
1365         enabled: true,
1366         items: [
1367             { id: 1, name: 'SMT 贴片产线', power: 350, count: 4 },
1368             { id: 2, name: '空压机组', power: 132, count: 3 }
1369         ]
1370     }
1371 }
1372 },
1373 {
1374     id: 2,
1375     name: '研发中心大楼',
1376     type: '办公楼 (Office)',
1377     area: 4800,
1378     systems: {
1379         lighting: {
1380             enabled: true,
1381             mode: 'density',
1382             density: 9,
1383             items: []
1384         },
1385         hvac: {
1386             enabled: true,
1387             items: [
1388                 { id: 1, name: 'VRV 多联机外机', type: '多联机 (VRF)', health: '优 (1-5年)', powe...
1389             ]
1390         },
1391         water: {
1392             enabled: true,
1393             items: [
1394                 { id: 1, name: '容积式电热水器', type: 'Heater', power: 15, count: 4 }
1395             ]
1396         },
1397         production: {
1398             enabled: false,
1399             items: []
1400         }
1401     }
```

```
1401   }
1402 },
1403 ];
1404
1405 const normalizeBuildingArea = (building: Building): Building => {
1406   const numericArea = Number(building.area);
1407   return {
1408     ...building,
1409     area: Number.isFinite(numericArea) ? numericArea : 0,
1410   };
1411 };
1412
1413 const ProjectEntry: React.FC = () => {
1414   const { transformers, setTransformers, saveProject, projectBaselInfo, setProjectBaselInfo...
1415
1416   // Region State (Hydrate from Context)
1417   const [projectName, setProjectName] = useState(projectBaselInfo.name);
1418   const [province, setProvince] = useState(projectBaselInfo.province);
1419   const [city, setCity] = useState(projectBaselInfo.city);
1420   const [projectType, setProjectType] = useState(projectBaselInfo.type);
1421
1422   // Building State (Hydrate from Context)
1423   const [buildings, setBuildings] = useState<Building[]>({
1424     (projectBaselInfo.buildings && projectBaselInfo.buildings.length > 0)
1425       ? projectBaselInfo.buildings.map(normalizeBuildingArea)
1426       : defaultBuildings
1427   });
1428
1429   const [targetBuildingId, setTargetBuildingId] = useState<number>(buildings.length > 0 ? ...
1430
1431   // Estimation Mode State
1432   const [estimationMode, setEstimationMode] = useState<'auto' | 'manual'>('manual');
1433   const [isScanning, setisScanning] = useState(false);
1434
1435   // System Config Tab State
1436   const [activeSystemTab, setActiveSystemTab] = useState<'lighting' | 'hvac' | 'water' | ...
1437
1438   // Sync target building if buildings change
1439   useEffect(() => {
1440     if (buildings.length > 0 && !buildings.find(b => b.id === targetBuildingId)) {
1441       setTargetBuildingId(buildings[0].id);
1442     }
1443   }, [buildings, targetBuildingId]);
1444
1445   // Sync to Global Context whenever local state changes
1446   useEffect(() => {
1447     setProjectBaselInfo({
1448       ...projectBaselInfo,
1449       name: projectName,
1450       type: projectType,
```

```
1451     province,
1452     city,
1453     buildings
1454   });
1455   }, [projectName, projectType, province, city, buildings, setProjectBaseInfo]);
1456
1457   // --- Handlers ---
1458
1459   const handleAddBuilding = useCallback(() => {
1460     setBuildings(prev => {
1461       const newId = prev.length > 0 ? Math.max(...prev.map(b => b.id)) + 1 : 1;
1462       const newBuilding: Building = {
1463         id: newId,
1464         name: `新建建筑 ${newId}`,
1465         type: '未定义',
1466         area: 1000,
1467         systems: {
1468           lighting: { enabled: true, mode: 'density', density: 10, items: [] },
1469           hvac: { enabled: true, items: [] },
1470           water: { enabled: false, items: [] },
1471           production: { enabled: false, items: [] }
1472         }
1473       };
1474       return [...prev, newBuilding];
1475     });
1476   }, []);
1477
1478   const handleDeleteBuilding = useCallback((id: number) => {
1479     setBuildings(prev => {
1480       const newBuildings = prev.filter(b => b.id !== id);
1481       if (targetBuildingId === id && newBuildings.length > 1) {
1482         setTargetBuildingId(newBuildings[0].id);
1483       }
1484       return newBuildings;
1485     });
1486   }, [targetBuildingId]);
1487
1488   const handleBuildingChange = useCallback((id: number, field: string, value: string | nu...
1489     const normalizedValue = field === 'area'
1490       ? (Number.isFinite(Number(value)) ? Number(value) : 0)
1491       : value;
1492     setBuildings(prev => prev.map(b => b.id === id ? { ...b, [field]: normalizedValue }...
1493   }, []);
1494
1495   // -- Deep Update Handlers --
1496
1497   const toggleSystemEnabled = useCallback((buildingId: number, systemKey: keyof Building[...
1498     setBuildings(prev => prev.map(b => {
1499       if (b.id !== buildingId) return b;
1500       return {
```

```
13621      { /* 方案名称 */}
13622      <div className="font-semibold text-[#1d1d1f] text-base mb-2 lea...
13623
13624      { /* 组件品牌 */}
13625      <div className="flex items-center gap-1.5 mb-3">
13626        <span className="material-icons text-[14px] text-slate-400"...
13627        <span className="text-xs text-slate-600">
13628          {solution.brand === 'longi' ? '隆基组件' : solution.brand =...
13629          {(solution.investmentMode || 'epc') === 'emc' ? ` ` : ${g...
13630        </span>
13631      </div>
13632      <div className="flex items-center gap-1.5 mb-3">
13633        <span className="material-icons text-[14px] text-slate-400"...
13634        <span className="text-xs text-slate-600">
13635          {CABLE_BRANDS[solution.cableBrand || 'generic'].name} · ...
13636        </span>
13637      </div>
13638      <div className="flex items-center gap-1.5 mb-3">
13639        <span className="material-icons text-[14px] text-slate-400"...
13640        <span className="text-xs text-slate-600">
13641          {SOLAR_CONSTRUCTION_METHODS[solution.constructionMethod...
13642        </span>
13643      </div>
13644
13645      { /* 建造成本单价展示 */}
13646      <div className="grid grid-cols-2 gap-2 pt-3 border-t border-sla...
13647      <div>
13648        <div className="text-[10px] text-slate-400">铺设容量</div>
13649        <div className="text-sm font-semibold text-slate-800">{...
13650      </div>
13651      <div>
13652        <div className="text-[10px] text-slate-400">建造成本</div>
13653        <div className="flex items-baseline gap-1">
13654          <span className="text-lg font-semibold text-[#0071e...
13655          <span className="text-[10px] text-slate-400">/Wp</s...
13656        </div>
13657      </div>
13658    </div>
13659
13660    { /* 选中指示器 */}
13661    {selectedSolutionId === solution.id && (
13662      <div className="absolute -top-2 -right-2 w-6 h-6 bg-[#0071e...
13663        <span className="material-icons text-white text-[14px]"...
13664      </div>
13665    )}
13666  </div>
13667  )}
13668 </div>
13669 })
13670
```

```

13671      /* 当前方案详细配置 */
13672      {currentSolution && (
13673          <div className="solar-apple-panel p-5 mt-4 space-y-5">
13674              <div className="grid grid-cols-1 md:grid-cols-2 gap-4">
13675                  <div className="bg-white/90 rounded-[18px] border border-white p-4 ...
13676                      <label className="text-xs font-semibold text-slate-500 mb-2 blo...
13677                      <div className="flex items-center gap-2">
13678                          <span className="material-icons text-slate-300 text-[18px]" ...
13679                          <input
13680                              type="text"
13681                              value={currentSolution.name}
13682                              onChange={(e) => onUpdateSolution(currentSolution.id, {...
13683                                  className="text-base font-semibold text-[#1d1d1f] bg-tr...
13684                                  placeholder="方案名称"
13685                              />
13686                      </div>
13687                  </div>
13688
13689                  <div className="bg-white/90 rounded-[18px] border border-white p-4 ...
13690                      <label className="text-xs font-semibold text-slate-500 mb-2 blo...
13691                      <div className="flex items-center gap-2">
13692                          <input
13693                              type="number"
13694                              step="1"
13695                              value={currentSolution.capacity ?? defaultCapacity}
13696                              onChange={(e) => {
13697                                  const nextCapacity = parseFloat(e.target.value) || 0;
13698                                  onUpdateSolution(currentSolution.id, { capacity: ne...
13699                              }}
13700                                  className="text-base font-semibold text-[#1d1d1f] bg-tr...
13701                              />
13702                      <span className="text-xs text-slate-400 font-mono">kWp</span>
13703                  </div>
13704                  <div className="mt-2 flex items-center justify-between gap-3">
13705                      <span className="text-[10px] text-slate-400">
13706                          {currentSolution.capacity ? '该方案使用独立容量' : '当前跟随楼栋容量汇总'}
13707                      </span>
13708                      {currentSolution.capacity && (
13709                          <button
13710                              onClick={() => onUpdateSolution(currentSolution.id,...
13711                              className="text-[10px] font-semibold text-[#0071e3]...
13712                          >
13713                              跟随楼栋汇总
13714                          </button>
13715                      )}
13716                  </div>
13717              </div>
13718          </div>
13719
13720          <div className="p-4 rounded-[18px] border border-white bg-white/90 shad...

```



```

13721 <label className="text-xs font-semibold text-slate-500 mb-2 block">...
13722 <div className="solar-apple-segment grid grid-cols-1 md:grid-cols-4...
13723   {[
13724     { id: 'emc', label: 'EMC 节能分成', icon: 'handshake' },
13725     { id: 'epc', label: 'EPC 工程总包', icon: 'construction' },
13726     { id: 'financing', label: '融资共建', icon: 'payments' },
13727     { id: 'co_build', label: '股权共建', icon: 'group_work' }
13728   ]}.map((mode) => (
13729     <button
13730       key={mode.id}
13731       onClick={() => onUpdateSolution(currentSolution.id, { i...
13732       className={ `solar-apple-segment-item flex items-center ...
13733     >
13734     <span className="material-icons text-[18px]">{mode.icon...
13735     {mode.label}
13736   </button>
13737   )]}
13738 </div>
13739 </div>
13740 </div>
13741   )}
13742 </div>
13743   );
13744 };
13745 // ===== FILE: modules/solar/index.tsx =====
13746 import React, { useMemo, useState } from 'react';
13747 import { useSolarRetrofit, useSolarMetrics, calculateSolarMetrics } from './hooks';
13748 import { SolarParamsState, SolarReportAudience } from './types';
13749 import { SolarForm } from './components/SolarForm';
13750 import { SolarCharts } from './components/SolarCharts';
13751 import SolarReport from './components/SolarReport';
13752 import { SolutionComparison } from './components/SolutionComparison';
13753 import { SolarFinancialDetails } from './components/SolarFinancialDetails';
13754 import { useProject } from '../context/ProjectContext';
13755 import { isModuleTakenOver } from '../utils/moduleAggregation';
13756 import { requestReportForModules } from '../shared/reporting';
13757 import { PvConsumptionChart } from '../shared/components/PvConsumptionChart';
13758 import { annualizeBillEnergy, buildPvConsumptionProfile } from '../shared/utils/pvConsum...
13759 import { getSolarProfileBasis } from '../shared/utils/solarGenerationProfile';
13760 import { optimizeSolarStorageCapacity } from '../shared/utils/solarStorageOptimization';
13761
13762 const RetrofitSolar: React.FC = () => {
13763   const { modules, updateModule } = useProject();
13764   const isTakenOver = isModuleTakenOver('retrofit-solar', modules);
13765   const {
13766     currentModule, params, handleUpdate, buildings, setBuildings,
13767     selfUseMode, setSelfUseMode, calculatedSelfConsumption, setCalculatedSelfConsumption,
13768     consumptionResult, toggleModule, saveProject, transformers, bills, projectBaseInfo,
13769     priceConfig, storageModule, sunHoursSource,
13770     // 新增：方案和品牌状态

```

```

13771   solutions, selectedSolutionId, currentSolution,
13772   handleSelectSolution, handleAddSolution, handleUpdateSolution, handleDeleteSolution
13773 } = useSolarRetrofit();
13774
13775 const { chartData, longTermMetrics } = useSolarMetrics(params, calculatedSelfConsumption);
13776 const billedEnergy = useMemo(() => annualizeBillEnergy(bills, {
13777   projectType: projectBaseInfo.type,
13778   province: projectBaseInfo.province,
13779   hasAirConditioning: projectBaseInfo.hasAirConditioning,
13780 }), [bills, projectBaseInfo.type, projectBaseInfo.province, projectBaseInfo.hasAirCondi...
13781 const pvConsumptionData = useMemo(() => {
13782   const storageParams = storageModule?.params || {};
13783   const villaAnnualLoadKwh = projectBaseInfo.type === 'villa'
13784     ? Math.max(0, Number(projectBaseInfo.villaDailyKwh || 0)) * 365
13785     : 0;
13786   const annualLoadKwh = villaAnnualLoadKwh
13787     || billedEnergy.annualizedKwh
13788     || transformers.reduce((total, transformer) => total + Number(transformer.capac...
13789     || 1000000;
13790   return buildPvConsumptionProfile({
13791     annualLoadKwh,
13792     projectType: projectBaseInfo.type,
13793     pvCapacityKw: Number(params.simpleParams.capacity || 0),
13794     dailySunHours: Number(params.advParams.dailySunHours || 4),
13795     performanceRatio: Number(params.advParams.prValue || 80) / 100 * Number(params....
13796     location: {
13797       latitude: projectBaseInfo.latitude,
13798       longitude: projectBaseInfo.longitude,
13799       province: projectBaseInfo.province,
13800       city: projectBaseInfo.city,
13801     },
13802     storage: {
13803       enabled: Boolean(storageModule?.isActive && storageParams.dispatchMode !== ...
13804       powerKw: Number(storageParams.basicParams?.power ?? 261),
13805       capacityKwh: Number(storageParams.basicParams?.capacity ?? 522),
13806       dod: Number(storageParams.advParams?.dod || 90) / 100,
13807       rte: Number(storageParams.advParams?.rte || 88) / 100,
13808     }
13809   });
13810 }, [billedEnergy.annualizedKwh, transformers, projectBaseInfo, params.simpleParams.capa...
13811 const solarProfileBasis = useMemo(() => getSolarProfileBasis(projectBaseInfo), [project...
13812 const hourlyPrices = useMemo(() => {
13813   if (priceConfig.mode === 'fixed') return Array(24).fill(Number(priceConfig.fixedPri...
13814   if (priceConfig.mode === 'spot') {
13815     return Array.from({ length: 24 }, (_, hour) => Number(priceConfig.spotPrices?.[...
13816   }
13817   return Array.from({ length: 24 }, (_, hour) => {
13818     const segment = priceConfig.touSegments.find(item => hour >= item.start && hour...
13819     return Number(segment?.price || priceConfig.fixedPrice || 0.85);
13820   });

```

```

13821   }, [priceConfig]);
13822   const annualLoadKwh = useMemo(() => {
13823     const villaAnnualLoadKwh = projectBaseInfo.type === 'villa'
13824       ? Math.max(0, Number(projectBaseInfo.villaDailyKwh | 0)) * 365
13825       : 0;
13826     return villaAnnualLoadKwh
13827       | | billedEnergy.annualizedKwh
13828       | | transformers.reduce((total, transformer) => total + Number(transformer.capac...
13829       | | 1000000;
13830   }, [projectBaseInfo.type, projectBaseInfo.villaDailyKwh, billedEnergy.annualizedKwh, tr...
13831   const storageParams = storageModule?.params | | {};
13832   const jointRecommendation = useMemo(() => optimizeSolarStorageCapacity({
13833     annualLoadKwh,
13834     projectType: projectBaseInfo.type,
13835     maxPvCapacityKw: Math.max(0, Number(params.simpleParams.capacity | 0)),
13836     dailySunHours: Number(params.advParams.dailySunHours | 4),
13837     performanceRatio: Number(params.advParams.prValue | 80) / 100
13838     * Number(params.advParams.azimuthEfficiency | 100) / 100,
13839     location: {
13840       latitude: projectBaseInfo.latitude,
13841       longitude: projectBaseInfo.longitude,
13842       province: projectBaseInfo.province,
13843       city: projectBaseInfo.city,
13844     },
13845     hourlyPrices,
13846     pvUnitCostYuanPerWp: Number(params.simpleParams.epcPrice | 2),
13847     storageUnitCostYuanPerKwh: Number(storageParams.basicParams?.unitCost | 1100),
13848     storageDod: Number(storageParams.advParams?.dod | 90) / 100,
13849     storageRte: Number(storageParams.advParams?.rte | 88) / 100,
13850     pvOmYuanPerWYear: Number(params.advParams.omCost | 0.03),
13851     storageOmRatePercent: Number(projectBaseInfo.omRate | 0),
13852     discountRatePercent: Number(projectBaseInfo.discountRate | 5),
13853     horizonYears: Number(params.advParams.projectLifeYears | 10),
13854     generationDays: Number(params.advParams.generationDays | 365),
13855     storageOperatingDays: 330,
13856   }), [
13857     annualLoadKwh, projectBaseInfo, params.simpleParams.capacity, params.simpleParams.e...
13858     params.advParams, hourlyPrices, storageParams,
13859   ]);
13860
13861   const [isChartExpanded, setIsChartExpanded] = useState(false);
13862   const [isFinancialModalOpen, setIsFinancialModalOpen] = useState(false);
13863   const [isSolarPresentationMode, setIsSolarPresentationMode] = useState(false);
13864   const [solarReportAudience, setSolarReportAudience] = useState<SolarReportAudience>('ow...
13865   const [isComparisonModalOpen, setIsComparisonModalOpen] = useState(false);
13866   const [selectedSolutionForDetails, setSelectedSolutionForDetails] = useState<string | n...
13867
13868   if (!currentModule) return null;
13869
13870   const applyJointRecommendation = () => {

```

```

13871   if (currentSolution) {
13872       handleUpdateSolution(currentSolution.id, { capacity: jointRecommendation.pvCapa...
13873   } else {
13874       handleUpdate({
13875           simpleParams: {
13876               ...params.simpleParams,
13877               capacity: jointRecommendation.pvCapacityKw,
13878           },
13879       });
13880   }
13881
13882   const nextStorageParams = {
13883       ...storageParams,
13884       dispatchMode: 'pv_surplus',
13885       basicParams: {
13886           ...(storageParams.basicParams || {}),
13887           power: jointRecommendation.storagePowerKw,
13888           capacity: jointRecommendation.storageCapacityKwh,
13889           unitCost: Number(storageParams.basicParams?.unitCost || 1100),
13890       },
13891       jointRecommendation,
13892   };
13893   updateModule('retrofit-storage', {
13894       isActive: jointRecommendation.storageRecommended,
13895       params: nextStorageParams,
13896       kpiPrimary: {
13897           label: '装机规模',
13898           value: jointRecommendation.storageRecommended
13899               ? `${jointRecommendation.storagePowerKw}kW/${jointRecommendation.storag...
13900               : '暂不配置',
13901       },
13902   });
13903 };
13904
13905 return (
13906     <div className="flex h-full bg-slate-50 relative">
13907         <div className="flex-1 flex flex-col min-w-0 overflow-hidden relative">
13908             <header className="h-16 bg-white border-b border-slate-200 flex items-cente...
13909                 <div className="flex items-center gap-4">
13910                     <div>
13911                         <h2 className="text-xl font-bold text-slate-900">分布式光伏配置</h2>
13912                         <p className="text-xs text-slate-500">屋顶光伏与BIPV一体化发电策略</p>
13913                     </div>
13914                     <div className="flex items-center gap-2 bg-slate-100 px-3 py-1.5 ro...
13915                         <span className={`text-xs font-bold ${currentModule.isActive ? ...
13916                             {currentModule.isActive ? '模块已启用' : '模块已停用'}
13917                         </span>
13918                         <button
13919                             onClick={() => toggleModule('retrofit-solar')}
13920                             className={`relative inline-flex h-5 w-9 items-center round...

```

```
13921         >
13922         <span className={`inline-block h-3 w-3 transform rounded-fu...
13923       </button>
13924     </div>
13925     {isTakenOver && <span className="text-[10px] font-bold text-indigo-...
13926   </div>
13927   <div className="flex items-center gap-3">
13928     <button onClick={() => requestReportForModules(['retrofit-solar'])}...
13929       <span className="material-icons text-base">summarize</span> 光伏独立汇报
13930   </button>
13931   <button
13932     onClick={() => requestReportForModules(['retrofit-solar', 'retr...
13933     disabled={!storageModule?.isActive}
13934     title={storageModule?.isActive ? '生成光伏与储能联合方案汇报' : '请先启用储能板块'}
13935     className="flex items-center gap-2 text-sm text-white font-bold...
13936   >
13937     <span className="material-icons text-base">account_tree</span> ...
13938   </button>
13939 </div>
13940 </header>
13941
13942 <div className={`flex-1 overflow-y-auto p-8 pb-32 transition-opacity durati...
13943 <div className="max-w-6xl mx-auto mb-6 rounded-2xl border-2 border-emer...
13944   <div className="flex flex-col lg:flex-row lg:items-start justif...
13945     <div className="flex-1">
13946       <div className="flex items-center gap-2 text-emerald-900">
13947         <span className="material-icons text-emerald-600">e...
13948         <h3 className="font-black">光储联合投资建议</h3>
13949         <span className="rounded-full bg-emerald-600 px-2 p...
13950         <span className="rounded-full bg-white/80 px-2 py-0...
13951           {params.simpleParams.operationMode === 'off_gri...
13952         </span>
13953       </div>
13954       <p className="mt-1 text-[11px] text-emerald-700">
13955         先按负荷确定经济光伏容量，再判断储能边际收益；储能不经济时直接建议为0。
13956         {params.simpleParams.operationMode !== 'off_grid' &...
13957       </p>
13958     </div>
13959     <button
13960       onClick={applyJointRecommendation}
13961       disabled={jointRecommendation.decision === 'defer'}
13962       className="shrink-0 rounded-xl bg-emerald-600 px-4 py-2...
13963   >
13964     一键应用联合建议
13965   </button>
13966 </div>
13967 <div className="mt-3 grid grid-cols-2 md:grid-cols-5 gap-3">
13968   <div><div className="text-[10px] text-emerald-600">建议光伏</di...
13969   <div><div className="text-[10px] text-emerald-600">建议储能</di...
13970   <div><div className="text-[10px] text-emerald-600">联合投资</di...
```

```
13971         <div><div className="text-[10px] text-emerald-600">首年净收益</d...
13972         <div><div className="text-[10px] text-emerald-600">静态回本</di...
13973     </div>
13974     <p className="mt-3 text-[11px] font-bold text-emerald-800">[joi...
13975     <div className="mt-3 grid grid-cols-2 md:grid-cols-4 gap-2 text...
13976         <div className="rounded-lg bg-white/70 px-3 py-2">项目期净现值：<s...
13977         <div className="rounded-lg bg-white/70 px-3 py-2">光伏消纳率：<st...
13978         <div className="rounded-lg bg-white/70 px-3 py-2">年光伏发电：<st...
13979         <div className="rounded-lg bg-white/70 px-3 py-2">年弃光：<stro...
13980     </div>
13981 </div>
13982 <SolarForm
13983     params={params}
13984     handleUpdate={handleUpdate}
13985     buildings={buildings}
13986     setBuildings={setBuildings}
13987     transformers={transformers}
13988     bills={bills}
13989     projectBaseInfo={projectBaseInfo}
13990     currentModule={currentModule}
13991     selfUseMode={selfUseMode}
13992     setSelfUseMode={setSelfUseMode}
13993     calculatedSelfConsumption={calculatedSelfConsumption}
13994     setCalculatedSelfConsumption={setCalculatedSelfConsumption}
13995     consumptionResult={consumptionResult}
13996     storageModule={storageModule}
13997     sunHoursSource={sunHoursSource}
13998     // 新增：方案和品牌相关
13999     solutions={solutions}
14000     selectedSolutionId={selectedSolutionId}
14001     currentSolution={currentSolution}
14002     handleSelectSolution={handleSelectSolution}
14003     handleAddSolution={handleAddSolution}
14004     handleUpdateSolution={handleUpdateSolution}
14005     handleDeleteSolution={handleDeleteSolution}
14006 />
14007 <div className="max-w-6xl mx-auto mt-6">
14008     <PvConsumptionChart
14009         data={pvConsumptionData}
14010         title="光伏发电与项目消纳曲线"
14011         dataBasis={` ${billedEnergy.monthCount} > 0
14012             ? `总用电量来自${billedEnergy.monthCount}个月真实账单并补齐全年；负荷形状按项目类型重构。`
14013             : `当前总用电量为估算值，导入账单后将自动归一化重算。`} 光伏曲线依据：${solarProfileBasis}及...
14014     />
14015 </div>
14016 </div>
14017
14018 { /* Sticky Footer */ }
14019 <div className="fixed bottom-0 left-64 right-[340px] bg-white/95 backdrop-b...
14020     <div className="flex items-center gap-3">
```

```
14021     <div className="w-8 h-8 rounded-full bg-slate-50 flex items-center ...
14022         <span className="material-icons text-[18px]">history</span>
14023     </div>
14024     <div className="flex flex-col">
14025         <span className="text-xs font-bold text-slate-700">自动同步</span>
14026         <span className="text-[10px] text-slate-400 font-medium">数据实时计算...
14027     </div>
14028 </div>
14029 <div className="flex items-center gap-3">
14030     <button className="px-6 py-2.5 text-sm font-semibold rounded-xl tex...
14031     <button
14032         onClick={saveProject}
14033         className="px-8 py-2.5 text-sm font-semibold rounded-xl bg-prim...
14034     >
14035         保存配置 <span className="material-icons text-[18px]">save</span>
14036     </button>
14037 </div>
14038 </div>
14039
14040 </div>
14041
14042 <aside className={`w-[340px] bg-white border-l border-slate-200 flex flex-col s...
14043 <div className="p-5 border-b border-slate-200 flex items-center justify-bet...
14044     <h3 className="font-bold text-slate-800 flex items-center gap-2">
14045         <span className="material-icons text-primary">analytics</span> 实时预估收益
14046     </h3>
14047     {!currentModule.isActive && <span className="text-xs font-bold text-red...
14048 </div>
14049
14050 <div className="flex-1 overflow-y-auto p-5 space-y-4 bg-slate-50/50">
14051     <div className="bg-white p-4 rounded-xl border border-slate-200 shadow-...
14052         <div className="flex items-center gap-2 mb-2">
14053             <div className="p-1.5 bg-yellow-100 rounded text-yellow-600"><s...
14054             <span className="text-xs font-semibold text-slate-500 uppercase...
14055         </div>
14056         <div className="flex items-baseline gap-2">
14057             <span className="text-3xl font-bold text-slate-900 tracking-tig...
14058         </div>
14059     </div>
14060
14061     <div className="bg-white p-4 rounded-xl border border-slate-200 shadow-...
14062         <div className="flex items-center gap-2 mb-2">
14063             <div className="p-1.5 bg-green-100 rounded text-green-600"><spa...
14064             <span className="text-xs font-semibold text-slate-500 uppercase...
14065         </div>
14066         <div className="flex items-baseline gap-2">
14067             <span className="text-3xl font-bold text-slate-900 tracking-tig...
14068             <span className="text-sm text-slate-500">万度</span>
14069         </div>
14070     </div>
```

```
14071
14072     <div className="bg-white p-4 rounded-xl border border-slate-200 shadow-...
14073         <div className="flex items-center gap-2 mb-2">
14074             <div className="p-1.5 bg-red-100 rounded text-red-600"><span cl...
14075             <span className="text-xs font-semibold text-slate-500 uppercase">
14076                 {params.simpleParams.investmentMode === 'emc'
14077                     ? '投资方首年净收益（税后）'
14078                     : params.simpleParams.investmentMode === 'co_build'
14079                     ? '业主首年综合收益'
14080                     : '首年净收益（税后）'}
14081             </span>
14082         </div>
14083         <div className="flex items-baseline gap-2">
14084             <span className="text-3xl font-bold text-slate-900 tracking-tig...
14085             <span className="text-sm text-slate-500">万元</span>
14086         </div>
14087     </div>
14088
14089     {params.simpleParams.investmentMode === 'emc' && (
14090         <div className="rounded-xl border border-amber-200 bg-gradient-to-b...
14091         <div className="mb-3 flex items-center justify-between">
14092             <div className="flex items-center gap-2">
14093                 <div className="rounded bg-amber-100 p-1.5 text-amber-7...
14094                 <span className="text-xs font-bold uppercase text-slate...
14095             </div>
14096             <span className="rounded-full bg-slate-900 px-2 py-1 text-[...
14097         </div>
14098         <div className="grid grid-cols-2 gap-3 text-sm">
14099             <div>
14100                 <p className="text-[10px] text-slate-400">初始投入</p>
14101                 <p className="mt-1 font-bold text-slate-900">¥ {Number{...
14102             </div>
14103             <div>
14104                 <p className="text-[10px] text-slate-400">投资方 IRR</p>
14105                 <p className="mt-1 font-bold text-blue-600">{Number(lon...
14106             </div>
14107             <div>
14108                 <p className="text-[10px] text-slate-400">回本周期</p>
14109                 <p className="mt-1 font-bold text-orange-600">{Number(l...
14110             </div>
14111             <div>
14112                 <p className="text-[10px] text-slate-400">{Math.max(1, ...
14113                 <p className="mt-1 font-bold text-emerald-600">¥ {Numbe...
14114             </div>
14115         </div>
14116     </div>
14117 }}
14118
14119     { /* Chart Container - Clickable */ }
14120     <div>
```



```
14121     className="bg-white p-4 rounded-xl border border-slate-200 shadow-s...
14122     onClick={() => setIsChartExpanded(true)}
14123   >
14124   <div className="flex items-center justify-between mb-3">
14125     <div className="flex items-center gap-2">
14126       <div className="p-1.5 bg-blue-100 rounded text-blue-600"><s...
14127       <span className="text-xs font-semibold text-slate-500 upper...
14128     </div>
14129     <span className="material-icons text-slate-300 text-sm group-ho...
14130   </div>
14131   <div className="h-24 w-full pointer-events-none flex items-end gap-1">
14132     {chartData.map(item => {
14133       const maxValue = Math.max(...chartData.map(row => Number(ro...
14134       return (
14135         <div key={item.name} className="flex-1 h-full flex item...
14136         <div
14137           className="w-full rounded-t-sm bg-amber-400"
14138           style={{ height: `${Math.max(4, Number(item.ret...
14139         />
14140       </div>
14141     );
14142   }}}
14143 </div>
14144   <div className="absolute inset-0 bg-white/0 group-hover:bg-white/5 ...
14145 </div>
14146
14147   { /* Financial Detail Trigger */ }
14148   <div
14149     onClick={() => setIsFinancialModalOpen(true)}
14150     className="p-4 bg-gradient-to-r from-slate-800 to-slate-700 rounded...
14151   >
14152     <div className="absolute right-0 top-0 w-16 h-16 bg-white/10 rounde...
14153     <div className="flex justify-between items-center relative z-10">
14154       <div>
14155         <h4 className="text-sm font-bold text-white flex items-cent...
14156         <span className="material-icons text-sm text-yellow-400...
14157       </h4>
14158       <p className="text-[10px] text-slate-300 mt-1">查看 {params.a...
14159     </div>
14160     <span className="material-icons text-white/50 group-hover:text-...
14161   </div>
14162 </div>
14163
14164   { /* Solar Report Trigger */ }
14165   <div
14166     onClick={(e) => {
14167       e.stopPropagation();
14168       setIsSolarPresentationMode(true);
14169     }}
14170     className="p-4 bg-gradient-to-r from-blue-600 to-indigo-700 rounded...
```

```
14171     >
14172     <div className="absolute right-0 top-0 w-16 h-16 bg-white/10 rounde...
14173     <div className="flex justify-between items-center relative z-10">
14174         <div>
14175             <h4 className="text-sm font-bold text-white flex items-cent...
14176                 <span className="material-icons text-sm text-yellow-400...
14177             </h4>
14178             <p className="text-[10px] text-blue-100 mt-1">
14179                 {params.simpleParams.investmentMode === 'emc'
14180                 ? `当前：${solarReportAudience === 'owner' ? '对业主汇报' : ...
14181                 : '点击查看项目收益评估PPT演示'}
14182             </p>
14183         </div>
14184         <span className="material-icons text-white/50 group-hover:text-...
14185     </div>
14186 </div>
14187
14188 {/* 方案对比触发按钮 */}
14189 <div
14190     onClick={(e) => {
14191         e.stopPropagation();
14192         setIsComparisonModalOpen(true);
14193     }}
14194     className="p-4 bg-gradient-to-r from-purple-600 to-indigo-700 round...
14195 >
14196     <div className="absolute right-0 top-0 w-16 h-16 bg-white/10 rounde...
14197     <div className="flex justify-between items-center relative z-10">
14198         <div>
14199             <h4 className="text-sm font-bold text-white flex items-cent...
14200                 <span className="material-icons text-sm text-yellow-400...
14201             </h4>
14202             <p className="text-[10px] text-purple-100">对比所有方案的财务指标</p>
14203         </div>
14204         <span className="material-icons text-white/50 group-hover:text-...
14205     </div>
14206 </div>
14207 </div>
14208 </aside>
14209
14210 <SolarCharts
14211     isChartExpanded={isChartExpanded}
14212     setIsChartExpanded={setIsChartExpanded}
14213     isFinancialModalOpen={isFinancialModalOpen}
14214     setIsFinancialModalOpen={setIsFinancialModalOpen}
14215     chartData={chartData}
14216     longTermMetrics={longTermMetrics}
14217     params={params}
14218     investment={currentModule.investment}
14219     handleUpdate={handleUpdate}
14220 />
```

```
14221
14222     {/* Solar Report - Direct Presentation Mode */}
14223     {isSolarPresentationMode && (
14224         <SolarReport
14225             onClose={() => setIsSolarPresentationMode(false)}
14226             defaultToPresentationMode={true}
14227             selfConsumptionRate={calculatedSelfConsumption}
14228             initialAudience={solarReportAudience}
14229             onAudienceChange={setSolarReportAudience}
14230         />
14231     )}
14232
14233     {/* 方案对比模态框 */}
14234     {isComparisonModalOpen && (
14235         <div
14236             className="fixed inset-0 bg-black/60 z-50 flex items-center justify-cen...
14237             onClick={() => {
14238                 setIsComparisonModalOpen(false);
14239                 setSelectedSolutionForDetails(null);
14240             }}
14241         >
14242             <div
14243                 className="bg-white rounded-3xl shadow-2xl w-full max-w-6xl max-h-[...
14244                 onClick={(e) => e.stopPropagation()}
14245             >
14246                 <div className="flex justify-between items-center px-8 py-6 border-...
14247                     <div className="flex items-center gap-4">
14248                         <div className="w-12 h-12 rounded-full bg-gradient-to-br fr...
14249                             <span className="material-icons text-white text-xl">com...
14250                         </div>
14251                         <div>
14252                             <h2 className="text-2xl font-bold text-slate-900">方案对比分...
14253                             <p className="text-sm text-slate-500">对比所有方案的技术参数和财务指标</p>
14254                         </div>
14255                     </div>
14256                     <button
14257                         onClick={() => {
14258                             setIsComparisonModalOpen(false);
14259                             setSelectedSolutionForDetails(null);
14260                         }}
14261                         className="w-10 h-10 flex items-center justify-center round...
14262                         title="关闭"
14263                     >
14264                         <span className="material-icons text-lg">close</span>
14265                     </button>
14266                 </div>
14267
14268                 <div className="flex-1 overflow-y-auto p-8 bg-slate-50/30">
14269                     <SolutionComparison
14270                         solutions={solutions}
```

```
14271         params={params}
14272         selfConsumptionRate={calculatedSelfConsumption}
14273     />
14274 </div>
14275 </div>
14276 </div>
14277   })
14278 </div>
14279   };
14280 };
14281
14282 export default RetrofitSolar;
14283 // ===== FILE: modules/solar/reporting/SolarReportChapter.tsx =====
14284 import React from 'react';
14285 import { Bar, CartesianGrid, ComposedChart, Legend, Line, ResponsiveContainer, Tooltip, XAx...
14286 import { calculateSolarMetrics } from '../hooks';
14287 import { CABLE_BRANDS, DEFAULTS, INVERTER_BRANDS, MODULE_BRANDS, type SolarParamsState } fr...
14288 import type { ModuleReportChapterProps } from '../../shared/reporting/chapters/types';
14289 import { ChapterHeader, ChapterPage, MetricCard, ReportFooter, SectionTitle } from '../../....
14290
14291 const modeLabel: Record<string, string> = { epc: '业主自投 (EPC)', emc: '合同能源管理 (EMC)', financing...
14292 const money = (value: number) => `¥${Number(value || 0).toFixed(2)}万`;
14293
14294 export const SolarReportChapter: React.FC<ModuleReportChapterProps> = ({ module, result, re...
14295   const params = { ...DEFAULTS, ...(module.params as Partial<SolarParamsState>), simplePara...
14296   const selfRate = Number(params.effectiveSelfConsumptionRate ?? report.counterfactuals.p?...
14297   const longTerm = calculateSolarMetrics(params, selfRate);
14298   const scenario = report.counterfactuals.p;
14299   const cashFlow = result.metrics.cashFlows.reduce<Array<{ year: number; annual: number; cu...
14300     rows.push({ year, annual: Number(value.toFixed(3)), cumulative: Number(((rows.at(-1)?c...
14301   return rows;
14302 }, []);
14303 const sensitivityRates = [...new Set([Math.max(40, selfRate - 15), selfRate, Math.min(100...
14304 const sensitivity = sensitivityRates.map(rate => {
14305   const metrics = calculateSolarMetrics(params, rate);
14306   return { rate: `${rate.toFixed(0)}%`, firstYear: Number(metrics.yearlyDetails?.[0]?.net...
14307   });
14308 const solutions = params.solutions || [];
14309 const selectedSolutionId = params.selectedSolutionId || solutions[0]?.id;
14310
14311 return <>
14312   <ChapterPage>
14313     <ChapterHeader number="02" icon="solar_power" title="分布式光伏完整方案" subtitle="推荐方案、工程配置、发...
14314     <div className="grid grid-cols-4 gap-3 mb-6">
14315       <MetricCard label="拟装机容量" value={` ${params.simpleParams.capacity.toFixed(1)} kWp` } ...
14316       <MetricCard label="首年发电量" value={` ${scenario?.annualPvGenerationKwh || 0} / 10000 ...
14317       <MetricCard label="光伏自用比例" value={` ${selfRate.toFixed(1)}%` } tone="emerald" />
14318       <MetricCard label="投资方式" value={modeLabel[params.simpleParams.investmentMode] || pa...
14319     </div>
14320     <SectionTitle icon="recommend">推荐方案配置</SectionTitle>
```

```
14321 <div className="grid grid-cols-2 gap-4 mb-6">
14322   {{
14323     ['安装面积', `${Number(params.simpleParams.area || 0).toLocaleString()} m²`,
14324     ['系统方式', params.simpleParams.operationMode === 'off_grid' ? '离网光伏 + 配套储能' : (params...
14325     ['建造单价', `${params.simpleParams.epcPrice.toFixed(2)} 元/Wp`],
14326     ['日均等效日照', `${params.advParams.dailySunHours.toFixed(2)} h/天`],
14327     ['系统效率 PR', `${params.advParams.prValue.toFixed(1)}%`],
14328     ['方位角效率', `${params.advParams.azimuthEfficiency.toFixed(1)}%`],
14329     ['上网电价', params.simpleParams.operationMode === 'off_grid' ? '不适用（余电不出售）' : `${par...
14330     ['项目地点', `${projectBaseInfo.province || ''} ${projectBaseInfo.city || ''}`],
14331   ].map(([label, value]) => <div key={label} className="rounded-xl border border-slat...
14332 </div>
14333 <SectionTitle icon="compare_arrows">备选方案对比</SectionTitle>
14334 <div className="overflow-hidden rounded-xl border border-slate-200">
14335   <table className="w-full text-xs"><thead className="bg-blue-600 text-white"><tr><th...
14336   <tbody className="divide-y divide-slate-100">{solutions.slice(0, 6).map(solution ...
14337     const capacity = Number(solution.capacity || params.simpleParams.capacity);
14338     const investment = capacity * Number(solution.epcPrice || params.simpleParams.e...
14339     return <tr key={solution.id} className={solution.id === selectedSolutionId ? 'b...
14340   }}</tbody>
14341 </table>
14342 </div>
14343 <ReportFooter label="光伏板块 · 方案与工程配置" />
14344 </ChapterPage>
14345
14346 <ChapterPage>
14347   <ChapterHeader number="02 · 2" icon="payments" title="光伏投资回报与消纳分析" subtitle="板块独立现金流口...
14348   <div className="grid grid-cols-4 gap-3 mb-6"><MetricCard label="光伏专项投资" value={money{...
14349   <SectionTitle icon="show_chart">25年年度收益与累计现金流</SectionTitle>
14350   <div className="h-[270px] rounded-xl border border-slate-200 bg-white p-3 mb-6"><Resp...
14351   <SectionTitle icon="timeline">消纳率敏感性</SectionTitle>
14352   <div className="h-[230px] rounded-xl border border-slate-200 bg-slate-50 p-3 mb-5"><R...
14353   <div className="rounded-xl bg-amber-50 border border-amber-200 p-4 text-xs text-amber...
14354   <ReportFooter label="光伏板块 · 收益与敏感性" />
14355 </ChapterPage>
14356 </>;
14357 };
14358
14359 export default SolarReportChapter;
14360 // ===== FILE: shared/reporting/chapters/registry.ts =====
14361 import SolarReportChapter from '../././modules/solar/reporting/SolarReportChapter';
14362 import StorageReportChapter from '../././modules/storage/reporting/StorageReportChapter';
14363 import type { ModuleReportChapterDefinition } from './types';
14364
14365 const chapterDefinitions: ModuleReportChapterDefinition[] = [
14366   { moduleId: 'retrofit-solar', title: '分布式光伏完整方案', component: SolarReportChapter },
14367   { moduleId: 'retrofit-storage', title: '工商业储能完整方案', component: StorageReportChapter },
14368 ];
14369
14370 export const moduleReportChapterRegistry = new Map(chapterDefinitions.map(definition => [de...
```

```
14371
14372 export const getModuleReportChapter = (moduleId: string) => moduleReportChapterRegistry.get...
14373 // ===== FILE: shared/reporting/chapters/types.ts =====
14374 import type React from 'react';
14375 import type { ModuleData } from '../..../context/ModuleContext';
14376 import type { ProjectBaseInfo } from '../..../context/ProjectContext';
14377 import type { CombinedReportResult, ModuleReportResult } from '../types';
14378
14379 export interface ModuleReportChapterProps {
14380   module: ModuleData;
14381   result: ModuleReportResult;
14382   report: CombinedReportResult;
14383   projectBaseInfo: ProjectBaseInfo;
14384 }
14385
14386 export interface ModuleReportChapterDefinition {
14387   moduleId: string;
14388   title: string;
14389   component: React.ComponentType<ModuleReportChapterProps>;
14390 }
14391 // ===== FILE: shared/reporting/financial.ts =====
14392 import { calculateIRR, calculatePaybackPeriod } from '../..../utils/financial';
14393 import type { FinancialMetrics } from '../types';
14394
14395 export const round3 = (value: number): number => Number((Number.isFinite(value) ? value : 0...
14396
14397 export const calculateNpv = (cashFlows: number[], discountRatePercent: number): number => {
14398   const rate = Math.max(-99, discountRatePercent) / 100;
14399   return round3(cashFlows.reduce((sum, flow, year) => sum + flow / Math.pow(1 + rate, year)...
14400   };
14401
14402 export interface CashFlowOptions {
14403   investment: number;
14404   firstYearBenefit: number;
14405   horizonYears: number;
14406   discountRate: number;
14407   annualDegradation?: number;
14408   annualOpex?: number;
14409   replacementYears?: number[];
14410   replacementCostRate?: number;
14411   residualRate?: number;
14412 }
14413
14414 export const buildFinancialMetrics = (options: CashFlowOptions): FinancialMetrics => {
14415   const {
14416     investment,
14417     firstYearBenefit,
14418     horizonYears,
14419     discountRate,
14420     annualDegradation = 0,
```

```

14421   annualOpex = 0,
14422   replacementYears = [],
14423   replacementCostRate = 0,
14424   residualRate = 0
14425 } = options;
14426 const cashFlows = [-Math.max(0, investment)];
14427 for (let year = 1; year <= horizonYears; year++) {
14428   const benefit = firstYearBenefit * Math.pow(1 - Math.max(0, annualDegradation) / 100, y...
14429   const replacement = replacementYears.includes(year) ? investment * replacementCostRate ...
14430   const residual = year === horizonYears ? investment * residualRate / 100 : 0;
14431   cashFlows.push(round3(benefit - annualOpex - replacement + residual));
14432 }
14433 return {
14434   investment: round3(investment),
14435   firstYearNetBenefit: round3(firstYearBenefit - annualOpex),
14436   npv: calculateNpv(cashFlows, discountRate),
14437   irr: round3(calculateIRR(cashFlows)),
14438   paybackPeriod: round3(calculatePaybackPeriod(cashFlows)),
14439   discountRate: round3(discountRate / 100),
14440   cashFlows
14441 };
14442 };
14443 // ===== FILE: shared/reporting/index.ts =====
14444 export * from './types';
14445 export * from './financial';
14446 export * from './adapters';
14447 export * from './physicalEngine';
14448 export * from './scenario';
14449 export * from './reportNavigation';
14450 // ===== FILE: shared/reporting/physicalEngine.ts =====
14451 import type { BuildReportInput, PhysicalScenarioResult, TypicalDayPoint } from './types';
14452 import { round3 } from './financial';
14453 import { getProjectLoadProfile } from '../utils/projectLoadProfiles';
14454 import { buildLocationSolarProfile } from '../utils/solarGenerationProfile';
14455 import { estimateAnnualLoad } from '../utils/monthlyLoadEstimation';
14456
14457 type PhysicalModule = 'solar' | 'storage' | 'ev';
14458 type Enabled = Record<PhysicalModule, boolean>;
14459
14460 const EV_BASE_PROFILE = [0.03, 0.02, 0.02, 0.02, 0.03, 0.05, 0.08, 0.12, 0.18, 0.21, 0.18, ...
14461
14462 const sum = (values: number[]) => values.reduce((total, value) => total + value, 0);
14463 const clamp = (value: number, min: number, max: number) => Math.min(max, Math.max(min, valu...
14464
14465 const pricesForDay = (input: BuildReportInput): number[] => {
14466   const config = input.priceConfig;
14467   if (config.mode === 'fixed') return Array(24).fill(Number(config.fixedPrice | 0));
14468   if (config.mode === 'spot') return Array.from({ length: 24 }, (_, hour) => Number(config....
14469   return Array.from({ length: 24 }, (_, hour) => {
14470     const segment = config.touSegments.find(item => hour >= item.start && hour < item.end);

```

```
14471   return Number(segment?.price || config.fixedPrice || 0.85);
14472 });
14473 };
14474
14475 const getSolar = (input: BuildReportInput) => {
14476   const solarModule = input.modules['retrofit-solar'];
14477   const params = solarModule?.params || {};
14478   const simple = params.simpleParams || {};
14479   const advanced = params.advParams || {};
14480   const capacityFromKpi = Number(String(solarModule?.kpiPrimary?.value || '').match(/\d+{?...
14481   return {
14482     capacity: Number(simple.capacity || capacityFromKpi),
14483     sunHours: Number(advanced.dailySunHours || 4),
14484     performance: Number(advanced.prValue || 80) / 100 * Number(advanced.azimuthEfficiency | ...
14485     feedInTariff: simple.operationMode === 'off_grid' ? 0 : Number(advanced.feedInTariff || ...
14486     isOffGrid: simple.operationMode === 'off_grid'
14487   };
14488 };
14489
14490 const getStorage = (input: BuildReportInput) => {
14491   const params = input.modules['retrofit-storage']?.params || {};
14492   const basic = params.basicParams || params;
14493   const advanced = params.advParams || params;
14494   return {
14495     power: Math.max(0, Number(basic.power ?? 261)),
14496     capacity: Math.max(0, Number(basic.capacity ?? 522)),
14497     dod: clamp(Number(advanced.dod || 90) / 100, 0.1, 1),
14498     rte: clamp(Number(advanced.rte || 88) / 100, 0.1, 1),
14499     demandPrice: Math.max(0, Number(params.economics?.demandPrice || params.demandPrice || ...
14500     dispatchMode: params.dispatchMode === 'hybrid' ? 'hybrid' : 'pv_surplus'
14501   };
14502 };
14503
14504 const getEv = (input: BuildReportInput) => {
14505   const params = input.modules['retrofit-ev']?.params || {};
14506   const quick = params.quickState || {};
14507   const precise = params.preciseState || {};
14508   let annualKwh = 0;
14509   let serviceRevenue = 0;
14510   if (params.mode === 'precise' && Array.isArray(precise.equipment)) {
14511     precise.equipment.forEach((item: any) => {
14512       const powers: Record<string, number> = { ac7: 7, dc60: 60, dc120: 120, v2g: 15 };
14513       const energy = Number(powers[item.type] || 7) * Number(item.count || 0) * Number(item...
14514       annualKwh += energy;
14515       serviceRevenue += energy * Number(item.serviceFee || 0);
14516     });
14517   } else {
14518     const power = Number(quick.acCount || 0) * 7 + Number(quick.dcCount || 0) * 60;
14519     annualKwh = power * Number(quick.turnover || 0) * 0.6 * 365;
14520     serviceRevenue = annualKwh * Number(quick.serviceFee || 0);
```



```
14521 }
14522 return { annualKwh, annualServiceRevenue: serviceRevenue };
14523 };
14524
14525 const buildEvProfile = (annualKwh: number, prices: number[], smart: boolean, solarEnabled: ...
14526   const dailyKwh = annualKwh / 365;
14527   const minPrice = Math.min(...prices);
14528   const weights = EV_BASE_PROFILE.map((base, hour) => {
14529     if (!smart) return base;
14530     const solarWeight = solarEnabled ? solarProfile[hour] * 1.8 : 0;
14531     const valleyWeight = prices[hour] <= minPrice * 1.05 ? 0.7 : 0;
14532     return base + solarWeight + valleyWeight;
14533   });
14534   const factor = dailyKwh / Math.max(0.0001, sum(weights));
14535   return weights.map(weight => weight * factor);
14536 };
14537
14538 export const simulatePhysicalScenario = (input: BuildReportInput, enabled: Enabled): Physic...
14539   const annualizedBillKwh = estimateAnnualLoad(input.bills, {
14540     projectType: input.projectBaseInfo.type,
14541     province: input.projectBaseInfo.province,
14542     hasAirConditioning: input.projectBaseInfo.hasAirConditioning,
14543   }).annualizedKwh;
14544   const villaAnnualLoadKwh = input.projectBaseInfo.type === 'villa'
14545     ? Math.max(0, Number(input.projectBaseInfo.villaDailyKwh || 0)) * 365
14546     : 0;
14547   const annualBaseKwh = villaAnnualLoadKwh
14548     || annualizedBillKwh
14549     || input.transformers.reduce((total, transformer) => total + Number(transformer.capacit...
14550     || 1000000;
14551   const baseDailyKwh = annualBaseKwh / 365;
14552   const baseProfile = getProjectLoadProfile(input.projectBaseInfo.type);
14553   const baseFactor = baseDailyKwh / sum(baseProfile);
14554   const baseLoad = baseProfile.map(value => value * baseFactor);
14555   const prices = pricesForDay(input);
14556   const solar = getSolar(input);
14557   const storage = getStorage(input);
14558   const ev = getEv(input);
14559   const pvDailyKwh = enabled.solar ? solar.capacity * solar.sunHours * solar.performance : 0;
14560   const locationSolarProfile = buildLocationSolarProfile({
14561     latitude: input.projectBaseInfo.latitude,
14562     longitude: input.projectBaseInfo.longitude,
14563     province: input.projectBaseInfo.province,
14564     city: input.projectBaseInfo.city,
14565   });
14566   const pvFactor = pvDailyKwh / Math.max(0.0001, sum(locationSolarProfile));
14567   const pvProfile = locationSolarProfile.map(value => value * pvFactor);
14568   const evProfile = enabled.ev ? buildEvProfile(ev.annualKwh, prices, enabled.storage || en...
14569   const lowPrice = [...prices].sort((a, b) => a - b)[Math.floor(prices.length * 0.3)];
14570   const highPrice = [...prices].sort((a, b) => a - b)[Math.floor(prices.length * 0.7)];
```

```
14571 const usableCapacity = enabled.storage ? storage.capacity * storage.dod : 0;
14572 const chargeEfficiency = Math.sqrt(storage.rte);
14573 const dischargeEfficiency = Math.sqrt(storage.rte);
14574 let soc = storage.dispatchMode === 'pv_surplus' ? 0 : usableCapacity * 0.15;
14575 const transformerLimit = input.transformers.reduce((total, transformer) => total + Number...
14576 const demandTarget = Math.min(transformerLimit, Math.max(...baseLoad) * 0.92);
14577 const rows: TypicalDayPoint[] = [];
14578
14579 for (let hour = 0; hour < 24; hour++) {
14580   const load = baseLoad[hour] + evProfile[hour];
14581   const pv = pvProfile[hour];
14582   let remainingLoad = Math.max(0, load - pv);
14583   let surplusPv = Math.max(0, pv - load);
14584   let storageCharge = 0;
14585   let storageDischarge = 0;
14586
14587   if (enabled.storage && surplusPv > 0 && soc < usableCapacity) {
14588     storageCharge = Math.min(storage.power, surplusPv, (usableCapacity - soc) / chargeEff...
14589     soc += storageCharge * chargeEfficiency;
14590     surplusPv -= storageCharge;
14591   }
14592   if (enabled.storage && remainingLoad > 0 && (storage.dispatchMode === 'pv_surplus' || p...
14593     const requiredForDemand = Math.max(0, remainingLoad - demandTarget);
14594     const desired = storage.dispatchMode === 'pv_surplus' || prices[hour] >= highPrice ? ...
14595     storageDischarge = Math.min(storage.power, desired, soc * dischargeEfficiency);
14596     soc -= storageDischarge / dischargeEfficiency;
14597     remainingLoad -= storageDischarge;
14598   }
14599   if (enabled.storage && storage.dispatchMode === 'hybrid' && prices[hour] <= lowPrice &&...
14600     const gridCharge = Math.min(storage.power, (usableCapacity - soc) / chargeEfficiency);
14601     storageCharge += gridCharge;
14602     soc += gridCharge * chargeEfficiency;
14603     remainingLoad += gridCharge;
14604   }
14605
14606   rows.push({
14607     hour,
14608     baseLoad: round3(baseLoad[hour]),
14609     evLoad: round3(evProfile[hour]),
14610     pvGeneration: round3(pv),
14611     storageCharge: round3(storageCharge),
14612     storageDischarge: round3(storageDischarge),
14613     gridImport: round3(Math.max(0, remainingLoad)),
14614     gridExport: round3(Math.max(0, surplusPv)),
14615     soc: round3(soc),
14616     price: round3(prices[hour])
14617   });
14618 }
14619
14620 const annualGridImportKwh = sum(rows.map(row => row.gridImport)) * 365;
```

```

14621 const annualGridExportKwh = sum(rows.map(row => row.gridExport)) * 365;
14622 const annualPvGenerationKwh = sum(rows.map(row => row.pvGeneration)) * 365;
14623 const annualStorageChargeKwh = sum(rows.map(row => row.storageCharge)) * 365;
14624 const annualStorageDischargeKwh = sum(rows.map(row => row.storageDischarge)) * 365;
14625 const annualEnergyCost = sum(rows.map(row => row.gridImport * row.price)) * 365;
14626 const peakGridKw = Math.max(...rows.map(row => row.gridImport));
14627 const annualDemandCost = peakGridKw * storage.demandPrice * 12;
14628 const annualFeedInRevenue = annualGridExportKwh * solar.feedInTariff;
14629 const annualEvServiceRevenue = enabled.ev ? ev.annualServiceRevenue : 0;
14630 const baselineCost = sum(baseLoad.map((value, hour) => value * prices[hour])) * 365
14631 + Math.max(...baseLoad) * storage.demandPrice * 12;
14632 const annualSystemValue = baselineCost - annualEnergyCost - annualDemandCost + annualFeed...
14633
14634 return {
14635   key: `${enabled.solar ? 'p': ''}${enabled.storage ? 's': ''}${enabled.ev ? 'e': ''}`...
14636   enabled,
14637   annualGridImportKwh: round3(annualGridImportKwh),
14638   annualGridExportKwh: round3(annualGridExportKwh),
14639   annualPvGenerationKwh: round3(annualPvGenerationKwh),
14640   annualEvEnergyKwh: round3(enabled.ev ? ev.annualKwh : 0),
14641   annualStorageChargeKwh: round3(annualStorageChargeKwh),
14642   annualStorageDischargeKwh: round3(annualStorageDischargeKwh),
14643   annualEnergyCost: round3(annualEnergyCost / 10000),
14644   annualDemandCost: round3(annualDemandCost / 10000),
14645   annualFeedInRevenue: round3(annualFeedInRevenue / 10000),
14646   annualEvServiceRevenue: round3(annualEvServiceRevenue / 10000),
14647   annualSystemValue: round3(annualSystemValue / 10000),
14648   peakGridKw: round3(peakGridKw),
14649   pvSelfConsumptionRate: annualPvGenerationKwh > 0 ? round3((1 - annualGridExportKwh / an...
14650   typicalDay: rows
14651 };
14652 };
14653
14654 export const buildCounterfactuals = (input: BuildReportInput): Record<string, PhysicalScena...
14655 const combinations: Enabled[] = [
14656   { solar: false, storage: false, ev: false },
14657   { solar: true, storage: false, ev: false },
14658   { solar: false, storage: true, ev: false },
14659   { solar: false, storage: false, ev: true },
14660   { solar: true, storage: true, ev: false },
14661   { solar: true, storage: false, ev: true },
14662   { solar: false, storage: true, ev: true },
14663   { solar: true, storage: true, ev: true }
14664 ];
14665 return Object.fromEntries(combinations.map(enabled => {
14666   const result = simulatePhysicalScenario(input, enabled);
14667   return [result.key, result];
14668 }));
14669 };
14670 // ===== FILE: shared/reporting/reportNavigation.ts =====

```

```
14671 const STORAGE_KEY = 'ZERO_CARBON_REPORT_SELECTION';
14672
14673 export const requestReportForModules = (moduleIds: string[]) => {
14674   sessionStorage.setItem(STORAGE_KEY, JSON.stringify(moduleIds));
14675   window.dispatchEvent(new CustomEvent('zero-carbon:open-report', { detail: { moduleIds } }));
14676 };
14677
14678 export const consumeRequestedReportModules = ((): string[] | null => {
14679   const raw = sessionStorage.getItem(STORAGE_KEY);
14680   if (!raw) return null;
14681   sessionStorage.removeItem(STORAGE_KEY);
14682   try {
14683     const value = JSON.parse(raw);
14684     return Array.isArray(value) ? value.filter(item => typeof item === 'string') : null;
14685   } catch {
14686     return null;
14687   }
14688 };
14689 // ===== FILE: shared/reporting/scenario.ts =====
14690 import { getReportAdapter } from './adapters';
14691 import { buildFinancialMetrics, round3 } from './financial';
14692 import { buildCounterfactuals } from './physicalEngine';
14693 import { getProjectTypeLabel } from './utils/projectLoadProfiles';
14694 import { estimateAnnualLoad } from './utils/monthlyLoadEstimation';
14695 import { getSolarProfileBasis } from './utils/solarGenerationProfile';
14696 import type {
14697   BuildReportInput,
14698   CombinedReportResult,
14699   InteractionValue,
14700   ParticipantLedger,
14701   PartyRole
14702 } from './types';
14703
14704 const PHYSICAL_IDS = ['retrofit-solar', 'retrofit-storage', 'retrofit-ev'];
14705 const toWan = (valueYuan: number) => valueYuan / 10000;
14706
14707 const participantName: Record<PartyRole, string> = {
14708   owner: '项目业主',
14709   solar_investor: '光伏投资方',
14710   storage_investor: '储能投资方',
14711   ev_operator: '充电运营方',
14712   financier: '融资方'
14713 };
14714
14715 const getMode = (input: BuildReportInput, moduleId: string): string => {
14716   const params = input.modules[moduleId]?params || {};
14717   if (moduleId === 'retrofit-solar') return params.simpleParams?.investmentMode || 'epc';
14718   if (moduleId === 'retrofit-storage') return params.investmentConfig?.mode || 'self';
14719   if (moduleId === 'retrofit-ev') return params.businessConfig?.mode || 'self_operated';
14720   return 'self';
```

```
14721 };
14722
14723 interface Allocation { investment: Partial<Record<PartyRole, number>>; benefit: Partial<Rec...
14724
14725 const moduleAllocation = (input: BuildReportInput, moduleId: string, investment: number, be...
14726   const params = input.modules[moduleId]?.params || {};
14727   const mode = getMode(input, moduleId);
14728   if (moduleId === 'retrofit-solar') {
14729     if (mode === 'emc') {
14730       const ownerRate = Number(params.advParams?.emcOwnerShareRate || 10) / 100;
14731       return { investment: { solar_investor: investment }, benefit: { owner: benefit * owne...
14732     }
14733     if (mode === 'co_build') {
14734       const investorRate = Number(params.advParams?.coBuildInvestorShareRate || 60) / 100;
14735       return { investment: { owner: investment * (1 - investorRate), solar_investor: invest...
14736     }
14737     return { investment: { owner: investment }, benefit: { owner: benefit }, assumptions: [...
14738   }
14739   if (moduleId === 'retrofit-storage') {
14740     if (mode === 'emc') {
14741       const ownerRate = Number(params.investmentConfig?.emcOwnerShareRate || 15) / 100;
14742       return { investment: { storage_investor: investment }, benefit: { owner: benefit * ow...
14743     }
14744     return { investment: { owner: investment }, benefit: { owner: benefit }, assumptions: [...
14745   }
14746   if (moduleId === 'retrofit-ev') {
14747     const config = params.businessConfig || {};
14748     if (mode === 'third_party') {
14749       const ownerRate = Number(config.ownerShareRate ?? 20) / 100;
14750       return { investment: { ev_operator: investment }, benefit: { owner: benefit * ownerRa...
14751     }
14752     if (mode === 'entrusted') {
14753       const commission = Number(config.operatorCommissionRate ?? 15) / 100;
14754       return { investment: { owner: investment }, benefit: { owner: benefit * (1 - commisi...
14755     }
14756     return { investment: { owner: investment }, benefit: { owner: benefit }, assumptions: [...
14757   }
14758   return { investment: { owner: investment }, benefit: { owner: benefit }, assumptions: [] };
14759 };
14760
14761 const allocateInteraction = (
14762   input: BuildReportInput,
14763   interaction: InteractionValue,
14764   benefits: Partial<Record<PartyRole, number>>,
14765   assumptions: string[]
14766 ) => {
14767   if (interaction.annualValue === 0) return;
14768   if (interaction.id.includes('storage') && getMode(input, 'retrofit-storage') === 'emc') {
14769     const ownerRate = Number(input.modules['retrofit-storage']?.params?.investmentConfig?.e...
14770     benefits.owner = (benefits.owner || 0) + interaction.annualValue * ownerRate;
```

```
14771   benefits.storage_investor = (benefits.storage_investor | | 0) + interaction.annualValue ...
14772   assumptions.push(`${interaction.label}按储能EMC分成规则结算。`);
14773   return;
14774 }
14775 if (interaction.id.includes('ev') && getMode(input, 'retrofit-ev') === 'third_party') {
14776   const ownerRate = Number(input.modules['retrofit-ev']?.params?.businessConfig?.ownerSha...
14777   benefits.owner = (benefits.owner | | 0) + interaction.annualValue * ownerRate;
14778   benefits.ev_operator = (benefits.ev_operator | | 0) + interaction.annualValue * (1 - own...
14779   assumptions.push(`${interaction.label}按充电运营分成规则结算。`);
14780   return;
14781 }
14782 benefits.owner = (benefits.owner | | 0) + interaction.annualValue;
14783 assumptions.push(`${interaction.label}未配置专项合同，默认归入业主节省。`);
14784 };
14785
14786 export const buildCombinedReport = (rawInput: BuildReportInput): CombinedReportResult => {
14787   const horizonYears = rawInput.horizonYears | | 25;
14788   const selectedModuleIds = rawInput.selectedModuleIds.filter(id => rawInput.modules[id]?i...
14789   const input = { ...rawInput, selectedModuleIds, horizonYears };
14790   let moduleResults = selectedModuleIds.map(moduleId => getReportAdapter(moduleId).buildSta...
14791     module: input.modules[moduleId],
14792     projectBaseInfo: input.projectBaseInfo,
14793     priceConfig: input.priceConfig,
14794     bills: input.bills,
14795     transformers: input.transformers,
14796     horizonYears
14797   });
14798   const counterfactuals = buildCounterfactuals(input);
14799   const hasSolar = selectedModuleIds.includes('retrofit-solar');
14800   const hasStorage = selectedModuleIds.includes('retrofit-storage');
14801   const hasEv = selectedModuleIds.includes('retrofit-ev');
14802   const billEstimation = estimateAnnualLoad(input.bills, {
14803     projectType: input.projectBaseInfo.type,
14804     province: input.projectBaseInfo.province,
14805     hasAirConditioning: input.projectBaseInfo.hasAirConditioning,
14806   });
14807   const billMonthCount = billEstimation.actualMonthCount;
14808   const storageParams = input.modules['retrofit-storage']?.params | | {};
14809   const storagesPvOnly = storageParams.dispatchMode !== 'hybrid';
14810   const storageHasValidScale = Number(storageParams.basicParams?.power ?? 261) > 0
14811     && Number(storageParams.basicParams?.capacity ?? 522) > 0;
14812   const physicalBenefitByModule: Record<string, number> = {
14813     'retrofit-solar': counterfactuals.p.annualSystemValue,
14814     'retrofit-storage': counterfactuals.s.annualSystemValue,
14815     'retrofit-ev': counterfactuals.e.annualSystemValue
14816   };
14817   moduleResults = moduleResults.map(result => {
14818     if (!result.moduleId in physicalBenefitByModule) return result;
14819     const degradation = result.moduleId === 'retrofit-solar'
14820       ? Number(input.modules[result.moduleId]?.params?.advParams?.degradationLinear | | 0.4)
```

```

14821 : result.moduleId === 'retrofit-storage'
14822 ? Number(input.modules[result.moduleId]?.params?.advParams?.degradation || 1.5)
14823 : 0.5;
14824 const replacementYears = result.moduleId === 'retrofit-storage' ? [12, 24] : result.mod...
14825 const replacementCostRate = result.moduleId === 'retrofit-storage' ? 60 : result.module...
14826 return {
14827   ...result,
14828   metrics: buildFinancialMetrics({
14829     investment: result.metrics.investment,
14830     firstYearBenefit: physicalBenefitByModule[result.moduleId],
14831     horizonYears,
14832     discountRate: Number(input.projectBaseInfo.discountRate || 5),
14833     annualDegradation: degradation,
14834     replacementYears,
14835     replacementCostRate,
14836     residualRate: result.moduleId === 'retrofit-solar' ? 5 : 0
14837   })
14838 };
14839 });
14840 const activeKey = `${hasSolar ? 'p' : ''}${hasStorage ? 's' : ''}${hasEv ? 'e' : ''}` || ...
14841 const physicalSingles = (hasSolar ? counterfactuals.p.annualSystemValue : 0)
14842 + (hasStorage ? counterfactuals.s.annualSystemValue : 0)
14843 + (hasEv ? counterfactuals.e.annualSystemValue : 0);
14844 const interactions: InteractionValue[] = [];
14845 if (hasSolar && hasStorage) interactions.push({ id: 'solar_storage', label: '光伏余电转存增量价值', ...
14846 if (hasSolar && hasEv) interactions.push({ id: 'solar_ev', label: '充电负荷提升光伏消纳价值', annualV...
14847 if (hasStorage && hasEv) interactions.push({ id: 'storage_ev', label: '储能降低充电需电价值', annua...
14848 if (hasSolar && hasStorage && hasEv) {
14849   interactions.push({
14850     id: 'solar_storage_ev',
14851     label: '光储充三方联合优化价值',
14852     annualValue: round3(counterfactuals.pse.annualSystemValue - counterfactuals.ps.annual...
14853   });
14854 }
14855 const otherResults = moduleResults.filter(result => !PHYSICAL_IDS.includes(result.moduleId...
14856 const otherAnnualBenefit = otherResults.reduce((total, result) => total + result.metrics....
14857 const activePhysical = counterfactuals[activeKey];
14858 const standaloneAnnualBenefit = round3(physicalSingles + otherAnnualBenefit);
14859 const interactionAnnualBenefit = round3(interactions.reduce((total, item) => total + item...
14860 const combinedAnnualBenefit = round3({ activeKey === 'base' ? 0 : activePhysical.annualSys...
14861 const totalInvestment = moduleResults.reduce((total, result) => total + result.metrics.in...
14862 const systemMetrics = buildFinancialMetrics({
14863   investment: totalInvestment,
14864   firstYearBenefit: combinedAnnualBenefit,
14865   horizonYears,
14866   discountRate: Number(input.projectBaseInfo.discountRate || 5),
14867   annualDegradation: 1,
14868   annualOpex: totalInvestment * Number(input.projectBaseInfo.omRate || 0) / 100
14869 });
14870

```

```

14871   const investments: Partial<Record<PartyRole, number>> = {};
14872   const benefits: Partial<Record<PartyRole, number>> = {};
14873   const ledgerAssumptions: Partial<Record<PartyRole, string[]>> = {};
14874   moduleResults.forEach(result => {
14875     const allocation = moduleAllocation(input, result.moduleId, result.metrics.investment, ...
14876     Object.entries(allocation.investment).forEach(([role, value]) => investments[role as Pa...
14877     Object.entries(allocation.benefit).forEach(([role, value]) => benefits[role as PartyRol...
14878     const targetRole = Object.keys(allocation.investment)[0] as PartyRole | | 'owner';
14879     ledgerAssumptions[targetRole] = [...(ledgerAssumptions[targetRole] | | []), ...allocatio...
14880   });
14881   const interactionAssumptions: string[] = [];
14882   interactions.forEach(interaction => allocateInteraction(input, interaction, benefits, int...
14883   ledgerAssumptions.owner = [...(ledgerAssumptions.owner | | []), ...interactionAssumptions];
14884
14885   const participantLedgers: ParticipantLedger[] = (Object.keys(participantName) as PartyRol...
14886   .filter(role => (investments[role] | | 0) !== 0 | | (benefits[role] | | 0) !== 0)
14887   .map(role => {
14888     const investment = Number(investments[role] | | 0);
14889     const benefit = Number(benefits[role] | | 0);
14890     return {
14891       id: role,
14892       role,
14893       name: participantName[role],
14894       investment: round3(investment),
14895       firstYearNetBenefit: round3(benefit),
14896       metrics: buildFinancialMetrics({ investment, firstYearBenefit: benefit, horizonYear...
14897       incomeSources: [{ label: '合同分配后首年净收益', value: round3(benefit) }],
14898       assumptions: ledgerAssumptions[role] | | []
14899     };
14900   });
14901
14902   const relationshipLabels = [
14903     ...(hasSolar && hasStorage ? [storagelsPvOnly ? '光伏 → 储能：余电转存与后续自用': '光伏 → 储能：余电转存与峰谷调...
14904     ...(hasSolar && hasEv ? ['光伏 → 充电桩：绿电直接消纳']: []),
14905     ...(hasStorage && hasEv ? ['储能 → 充电桩：需量保护与错峰供能']: [])
14906   ];
14907   const unsupportedCombination = selectedModuleIds.length > 1 && relationshipLabels.length ...
14908   const dataQuality = moduleResults.every(result => result.dataQuality === 'measured') ? 'm...
14909   return {
14910     scenarioName: selectedModuleIds.length === 1 ? `${moduleResults[0]?.name} | | '单板块'独立汇报`...
14911     selectedModuleIds,
14912     generatedAt: new Date().toISOString(),
14913     dataQuality,
14914     modules: moduleResults,
14915     activePhysicalScenario: activeKey === 'base' ? undefined : activePhysical,
14916     counterfactuals,
14917     interactions,
14918     standaloneAnnualBenefit,
14919     interactionAnnualBenefit,
14920     combinedAnnualBenefit,

```



```
14921   systemMetrics,
14922   participantLedgers,
14923   relationshipLabels,
14924   warnings: [
14925     ...[unsupportedCombination ? ['当前选择的板块之间尚未配置物理联动模型，本报告仅汇总独立财务结果。']: []],
14926     ...[hasSolar && activePhysical.annualPvGenerationKwh <= 0 ? ['光伏板块缺少有效装机或发电参数，未计算光伏物理...
14927     ...[hasStorage && !storageHasValidScale ? ['储能板块缺少有效功率/容量参数，未计算储能物理收益。']: []],
14928     ...[hasStorage && storagelsPvOnly && !hasSolar ? ['当前储能采用光伏余电专用策略；未选择光伏板块时，储能独立场景不产生充...
14929     ...[hasStorage && storagelsPvOnly && hasSolar && activePhysical.annualStorageChargeKw...
14930     ...[hasEv && activePhysical.annualEvEnergyKwh <= 0 ? ['充电桩板块缺少有效设备或利用率参数，未计算充电运营收益。']...
14931     ...[dataQuality === 'estimated' ? ['至少一个板块使用快速估算数据，正式投资决策前应使用实测负荷复核。']: []],
14932     ...moduleResults.flatMap(result => result.warnings)
14933   ],
14934   assumptions: [
14935     '系统经济账只统计项目边界外部现金流，参与方之间的内部结算在合并时抵消。',
14936     '光储充协同收益来自反事实场景差额，不使用固定增益比例。',
14937     ...[billMonthCount > 0 ? ['已有${billMonthCount}个月真实电费数据；缺失的${billEstimation.estimatedM...
14938     ...[hasSolar ? ['光伏典型日出力形状按${getSolarProfileBasis(input.projectBaseInfo)}计算，并以当地等效日照小...
14939     ...[hasStorage && storagelsPvOnly ? ['储能按${Number(storageParams.basicParams?.power ??...
14940     ...[hasStorage && Number(storageParams.recommendation?.capacity | | 0) > 0 ? ['基于${sto...
14941     `联合财务测算期为${horizonYears}年。`
14942   ]
14943 };
14944 };
14945
14946 export const formatWan = (value: number) => `¥${round3(value).toLocaleString('zh-CN', { min...
14947 export const formatKwh = (value: number) => `${round3(value / 10000).toLocaleString('zh-CN'...
14948 export const yuanToWan = toWan;
14949 // ===== FILE: shared/reporting/types.ts =====
14950 import type { ModuleData } from '../..context/ModuleContext';
14951 import type { Bill, PriceConfigState, Transformer } from '../..context/ConfigContext';
14952 import type { ProjectBaseInfo } from '../..context/ProjectContext';
14953
14954 export type ReportModuleId = ModuleData['id'];
14955 export type DataQuality = 'estimated' | 'measured';
14956 export type PartyRole = 'owner' | 'solar_investor' | 'storage_investor' | 'ev_operator' | '...
14957
14958 export interface FinancialMetrics {
14959   investment: number;
14960   firstYearNetBenefit: number;
14961   npv: number;
14962   irr: number;
14963   paybackPeriod: number;
14964   /** Decimal form used for display/export. e.g. 0.05 means 5%. */
14965   discountRate: number;
14966   cashFlows: number[];
14967 }
14968
14969 export interface TechnicalMetric {
14970   label: string;
```

```
14971   value: string;
14972 }
14973
14974 export interface ModuleReportResult {
14975   moduleId: string;
14976   name: string;
14977   strategy: string;
14978   investmentMode: string;
14979   dataQuality: DataQuality;
14980   metrics: FinancialMetrics;
14981   technicalMetrics: TechnicalMetric[];
14982   assumptions: string[];
14983   warnings: string[];
14984 }
14985
14986 export interface ReportAdapterContext {
14987   module: ModuleData;
14988   projectBaseInfo: ProjectBaseInfo;
14989   priceConfig: PriceConfigState;
14990   bills: Bill[];
14991   transformers: Transformer[];
14992   horizonYears: number;
14993 }
14994
14995 export interface ModuleReportAdapter {
14996   moduleId: string;
14997   capabilities: Array<'generation' | 'storage' | 'flexible_load' | 'load_reduction' | 'fina...
14998   buildStandalone(context: ReportAdapterContext): ModuleReportResult;
14999 }
15000
15001 export interface TypicalDayPoint {
15002   hour: number;
15003   baseLoad: number;
15004   evLoad: number;
15005   pvGeneration: number;
15006   storageCharge: number;
15007   storageDischarge: number;
15008   gridImport: number;
15009   gridExport: number;
15010   soc: number;
15011   price: number;
15012 }
15013
15014 export interface PhysicalScenarioResult {
15015   key: string;
15016   enabled: { solar: boolean; storage: boolean; ev: boolean };
15017   annualGridImportKwh: number;
15018   annualGridExportKwh: number;
15019   annualPvGenerationKwh: number;
15020   annualEvEnergyKwh: number;
```

```
15021   annualStorageChargeKwh: number;
15022   annualStorageDischargeKwh: number;
15023   annualEnergyCost: number;
15024   annualDemandCost: number;
15025   annualFeedInRevenue: number;
15026   annualEvServiceRevenue: number;
15027   annualSystemValue: number;
15028   peakGridKw: number;
15029   pvSelfConsumptionRate: number;
15030   typicalDay: TypicalDayPoint[];
15031 }
15032
15033 export interface InteractionValue {
15034   id: 'solar_storage' | 'solar_ev' | 'storage_ev' | 'solar_storage_ev';
15035   label: string;
15036   annualValue: number;
15037 }
15038
15039 export interface ParticipantLedger {
15040   id: PartyRole;
15041   name: string;
15042   role: PartyRole;
15043   investment: number;
15044   firstYearNetBenefit: number;
15045   metrics: FinancialMetrics;
15046   incomeSources: Array<{ label: string; value: number }>;
15047   assumptions: string[];
15048 }
15049
15050 export interface CombinedReportResult {
15051   scenarioName: string;
15052   selectedModuleIds: string[];
15053   generatedAt: string;
15054   dataQuality: DataQuality;
15055   modules: ModuleReportResult[];
15056   activePhysicalScenario?: PhysicalScenarioResult;
15057   counterfactuals: Record<string, PhysicalScenarioResult>;
15058   interactions: InteractionValue[];
15059   standaloneAnnualBenefit: number;
15060   interactionAnnualBenefit: number;
15061   combinedAnnualBenefit: number;
15062   systemMetrics: FinancialMetrics;
15063   participantLedgers: ParticipantLedger[];
15064   relationshipLabels: string[];
15065   warnings: string[];
15066   assumptions: string[];
15067 }
15068
15069 export interface BuildReportInput {
15070   selectedModuleIds: string[];
```

```
15071 modules: Record<string, ModuleData>;
15072 projectBaseInfo: ProjectBaseInfo;
15073 priceConfig: PriceConfigState;
15074 bills: Bill[];
15075 transformers: Transformer[];
15076 horizonYears?: number;
15077 }
15078 // ===== FILE: shared/reporting/chapters/ChapterLayout.tsx =====
15079 import React from 'react';
15080 import { PRODUCT_IDENTITY } from '../config/productIdentity';
15081
15082 export const ChapterPage: React.FC<{ children: React.ReactNode; className?: string }> = ({ ...
15083   <section className={`p-[14mm] min-h-[297mm] bg-white page-break-inside-avoid ${className}`}>
15084     {children}
15085     <div className="page-break" />
15086   </section>
15087 );
15088
15089 export const ChapterHeader: React.FC<{ number: string; icon: string; title: string; subtitl...
15090   const classes = tone === 'emerald'
15091     ? 'from-emerald-600 to-teal-700 text-emerald-100'
15092     : 'from-blue-600 to-indigo-700 text-blue-100';
15093   return (
15094     <header className={`bg-gradient-to-r ${classes} text-white rounded-2xl p-6 mb-6 shadow-...
15095       <div className="flex items-center gap-4">
15096         <div className="w-12 h-12 rounded-xl bg-white/15 flex items-center justify-center">...
15097         <div><div className="text-[10px] font-bold tracking-[0.25em] opacity-80">CHAPTER {n...
15098       </div>
15099     </header>
15100   );
15101 };
15102
15103 export const SectionTitle: React.FC<{ icon: string; children: React.ReactNode }> = ({ icon,...
15104   <h3 className="text-lg font-black text-slate-900 mb-4 pb-3 border-b border-slate-100 flex...
15105 );
15106
15107 export const MetricCard: React.FC<{ label: string; value: string; tone?: 'slate' | 'blue' | ...
15108   const tones = {
15109     slate: 'from-slate-50 to-slate-100 border-slate-200 text-slate-800',
15110     blue: 'from-blue-50 to-blue-100 border-blue-100 text-blue-700',
15111     emerald: 'from-emerald-50 to-green-100 border-emerald-100 text-emerald-700',
15112     violet: 'from-violet-50 to-purple-100 border-violet-100 text-violet-700',
15113     amber: 'from-amber-50 to-orange-100 border-amber-100 text-amber-700',
15114   }[tone];
15115   return <div className={`rounded-xl border bg-gradient-to-br p-4 ${tones}`}><div className...
15116 );
15117
15118 export const ReportFooter: React.FC<{ label: string }> = ({ label }) => (
15119   <footer className="mt-7 pt-3 border-t border-slate-100 flex justify-between text-[9px] te...
15120 );
```